

Autoregressive Models for High-Frequency Financial Data

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Summary

This dissertation examines the time between financial market events and how these durations depend on recent trading activity. It uses tick-level data for GBP/USD foreign exchange quote changes, BTCUSDT trades on Binance, and NVDA equity trades over 1-23 May 2026. These assets represent three different market structures: decentralised foreign exchange, continuously traded cryptocurrency, and an equity market with fixed trading hours.

The data are processed carefully before modelling. Non-price-changing GBP/USD quotes are removed, NVDA observations are restricted to regular trading hours, and records with identical timestamps are consolidated. Nanosecond timestamps are stored using integer components to avoid incorrectly merging distinct events. Durations are calculated only within continuous trading sessions, preventing market closures from creating artificial waiting times. Intraday seasonality is estimated from the training data using a smoothed five-minute duration profile and removed before model fitting.

The adjustment reduces the Ljung–Box statistic by 50.6% for GBP/USD and 53.7% for NVDA, showing that daily trading patterns explain a substantial part of their duration dependence. The reduction is only 4.5% for BTCUSDT, consistent with its continuous 24-hour market. Strong dependence nevertheless remains in all three adjusted series. The homogeneous Poisson model is rejected for every asset because observed duration variability is approximately two to five times greater than the model permits.

Exponential, Weibull, and two logarithmic ACD specifications are compared using residual diagnostics, AIC, and BIC. The Weibull ACD(2,2) model provides the strongest fit for GBP/USD, with persistence of 0.9993. For BTCUSDT and NVDA, the Exponential ACD(2,2) model produces the lowest squared-residual Ljung–Box statistics, although its persistence estimates are slightly above one. The Weibull model is unreliable for these two assets because its fitted density is strongly affected by their concentration of durations near zero.

The two Log-ACD formulations produce clearly different results. LACD1 leaves substantial residual autocorrelation and does not adequately capture duration dependence. LACD2 performs considerably better, with the (2,2) order preferred for all three assets within that model family. Its estimated persistence is close to, but below, one. However, persistence in a Log-ACD model is defined on the logarithmic conditional-duration scale and cannot be compared directly with the linear ACD persistence measure.

Additional checks show that the above-unity linear persistence estimates for BTCUSDT and NVDA are strongly influenced by very short, closely spaced trades. Merging these observations or reconstructing events using exchange information moves persistence towards or below one, while reducing the sample size alone has little effect. Overall, the findings show that ACD models remain useful, but reliable interpretation requires careful treatment of timestamp precision, event construction, intraday patterns, and near-simultaneous trades.

Declaration: Use of Generative AI

Name & version	Publisher	URL	How it was used (one sentence)
Claude	Anthropic	https://claude.ai/	Used for coding assistance, error correction, and checks of spelling and grammar.

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Chapter 1

Introduction and Literature Review

1.1 Motivation

Advances in electronic trading have made financial transaction data available at increasingly fine time resolutions, including milliseconds, microseconds, and nanoseconds. A single asset can therefore generate millions of observations in a relatively short period. In such data, the timing of events is informative rather than simply a label attached to prices. Periods of rapid trading tend to be followed by further rapid activity, while quieter periods tend to remain quiet. This clustering is not simply explained by the familiar intraday pattern in financial markets, since dependence can remain after this deterministic component is removed. Modelling the time between events therefore provides a direct way to study the dynamics of trading activity.

1.2 The Duration as the Object of Study

The variable of interest is the *duration*, defined as the time between consecutive events:

$$x_i = t_i - t_{i-1}, \quad (1.1)$$

where t_i is the time of event i . Short durations indicate rapid trading activity, while longer durations indicate quieter periods. Unlike regularly sampled financial data, transaction data do not arrive at fixed intervals, making the time between events itself an important object of study.

1.3 The Autoregressive Conditional Duration Framework

Conventional time-series models are designed mainly for equally spaced observations and are therefore not naturally suited to transaction data. Durations are also strictly positive and their variability changes with the level of activity. The Autoregressive Conditional Duration (ACD) model introduced by Engle and Russell [7] was developed specifically for this setting. Instead of imposing a regular time grid, the model directly describes the conditional expected duration

given the history of the process:

$$E(x_i | x_{i-1}, \dots, x_1) = \psi_i(x_{i-1}, \dots, x_1; \theta) = \psi_i. \quad (1.2)$$

The ACD model is written as

$$x_i = \psi_i \varepsilon_i, \quad (1.3)$$

where

$$\varepsilon_i \sim i.i.d. \quad (1.4)$$

is a positive innovation. This structure guarantees positive durations and allows the conditional mean to depend on previous durations, capturing clustering in trading activity. The framework is closely related to the GARCH model of Bollerslev [4], with durations playing a role analogous to volatility.

1.4 Literature Review

1.4.1 Development and Extensions of ACD Models

The ACD framework was introduced by Engle and Russell [7], following related work on foreign exchange quote durations by Engle and Russell [6]. These studies established that financial durations exhibit serial dependence, with periods of fast and slow activity tending to cluster.

The original model used an exponential innovation distribution because of its simplicity and convenient estimation properties. However, its constant hazard rate can be restrictive. Engle and Russell [7] found evidence supporting the Weibull distribution, while Lunde [10] introduced the generalized gamma specification, which includes the exponential and Weibull distributions as special cases. Grammig and Maurer [8] proposed the Burr distribution to allow more flexible hazard behaviour, and Vuorenmaa [15] considered the q-Weibull distribution.

Extensions have also focused on the conditional mean. Bauwens and Giot [1] proposed the logarithmic ACD (LACD) model, which models the logarithm of the conditional mean and guarantees positive durations without requiring non-negative parameter restrictions. This logarithmic form has since been developed further, for example by Tan et al. [14], who combine it with a component structure and a flexible error distribution.

1.4.2 Very Short Durations and High-Frequency Data

The increasing resolution of transaction data introduces an additional issue: very short and zero-duration observations can arise because events are recorded at finite timestamp resolutions. Blasques et al. [3] highlight the difference between the continuous durations assumed by conventional ACD models and the discrete nature of recorded transaction times.

This is particularly relevant for modern electronic markets, where microsecond and finer timestamps can produce large numbers of very short durations. Nakakita and Nakatsuma [11]

study trading interval durations in the Bitcoin market using high-frequency transaction data and report the same concentration of very short durations seen here. Such observations may affect the estimated persistence and therefore the interpretation of the fitted ACD process. This motivates examining the sensitivity of ACD estimates to the treatment of short durations in the present study.

1.4.3 Estimation, Diagnostics and Persistence

ACD models are generally estimated by maximum likelihood, with the exponential specification providing a convenient quasi-maximum-likelihood baseline [7]. Model adequacy is commonly assessed using standardised residuals and tests for remaining serial dependence, including the Ljung–Box statistic applied to residuals and their squares. Pacurar [13] reviews these methods and notes that the very large samples typical of transaction data can make diagnostic tests highly sensitive to small amounts of remaining dependence.

Persistence is another important feature of ACD models and is determined by the sum of the autoregressive coefficients in standard specifications. Values close to or above one require care because the usual stationarity interpretation becomes less straightforward. Cavaliere et al. [5] develop asymptotic results for ACD models in this near-unit-root and integrated regime.

Previous empirical research has documented substantial persistence in intertrade durations. Jasiak [9] found slowly decaying autocorrelations in transaction durations and investigated this persistence using a fractionally integrated autoregressive conditional duration model.

1.5 Structure of the Dissertation

Chapter 2 describes the three datasets and explains how the duration series are constructed from the raw transaction data, including timestamp handling, event filtering, and intraday adjustment.

Chapter 3 presents the ACD specifications, estimation results, model diagnostics, and empirical comparisons, including the sensitivity of persistence to very short durations.

The dissertation concludes by discussing the implications of the findings for ACD modelling of modern high-frequency financial data.

Chapter 2

Data and Duration Construction

This chapter describes the three datasets, the data pre-processing steps to convert event times into durations and diurnal adjustment of durations.

2.1 The Three Assets

This dissertation uses three datasets covering a common period of one month, from 1 May 2026 to 23 May 2026. The number of trading days differs across the three assets according to the operating schedule of each market, and also the number of events. Three different assets with varying market structure and events were chosen. Their differences in trading mechanism, operating schedule, and the nature of the recorded event are set out below.

2.1.1 Foreign Exchange: GBP/USD

The foreign exchange (forex) data consist of quote records for the GBP/USD currency pair, obtained from Dukascopy and retrieved programmatically before being stored locally in comma-separated format. Each record comprises a timestamp indicating the instant at which the quote was published, an ask price giving the rate at which the counterparty is willing to sell, and a bid price giving the rate at which it is willing to buy. The mid-price, defined as the mean of the bid and ask, is used throughout as the reference price, since it is not subject to the bid-ask bounce that affects transaction prices. Foreign exchange is traded over the counter rather than on a centralised exchange. There is no matching engine and no consolidated tape, and executed trades are not publicly reported. The observable record therefore consists of quote revisions rather than transactions. The market operates continuously from 10:00 (UTC) on Sunday to 17:00 (UTC) on Friday, closing over the weekend. Here, the event is the revision of a quoted price. This follows Engle and Russell [6], who apply the same framework to foreign exchange quote durations for the same reason.

2.1.2 Cryptocurrency: BTCUSDT

The cryptocurrency data consist of aggregated trade records for the BTCUSDT pair on the Binance exchange, retrieved from the exchange’s public data archive. Binance publishes trades in aggregated rather than raw form: consecutive fills arising from a single taker order executing against multiple resting orders at the same price and instant are combined into one record, comprising a trade identifier, the execution price, the aggregated quantity, the identifiers of the first and last constituent trades (which together delimit the range of raw trades the record represents), the timestamp, and flags indicating whether the buyer was the maker and whether the trade occurred at the best available price. Because this upstream aggregation is only partial, a substantial proportion of records still share a timestamp with their neighbours. Unlike the other two markets, the exchange operates continuously, with no closures, weekends, or auction periods, and the event of interest is the exchange-reported trade.

2.1.3 Equity (Stock): NVDA

The equity data consist of market-by-order records for NVIDIA Corporation common stock, obtained from Databento at nanosecond resolution. Each record comprises the execution price, the fill size, an action code identifying the event type from which trade events are selected, a set of flags describing the state of the record (including whether it terminates a batch of fills sharing a timestamp), an exchange-assigned sequence number, and the event timestamp in nanoseconds. The market trades five days a week within a defined session, subject also to opening and closing auctions from 13:30 to 20:00 that behave differently from continuous trading. As with BTCUSDT, the event of interest is the exchange-reported trade.

2.2 Constructing the Duration Series

The three datasets differ in the source, resolution, and nature of the recorded event. In this section the available data are converted into a common form across the three assets, suitable for duration calculation and modelling. Each asset is handled differently because the available resolution differs, but the end result is the same in every case: one duration per economically distinct arrival event.

2.2.1 Timestamp Precision

R’s `POSIXct` stores timestamps as seconds since 1 January 1970 using double precision floating-point numbers. At current dates, this leaves insufficient precision to represent nanosecond timestamps exactly. As a result, distinct trades that occur only nanoseconds apart can be rounded to the same timestamp. This is particularly problematic because the consolidation step in Section 2.2.3 treats simultaneous events as a single arrival, potentially merging genuinely separate trades.

To avoid this, timestamps are stored as two integers: whole seconds since the Unix epoch and the nanosecond remainder within the second. Durations are then calculated before converting to floating-point form:

$$x_i = (s_i - s_{i-1}) + \frac{f_i - f_{i-1}}{10^9}, \quad (2.1)$$

where s and f are the seconds and nanosecond-remainder components, respectively. This preserves the original timestamp information and avoids artificial zero or near-zero durations, which is particularly important for the NVDA data.

Later, all three sources are mapped to a common schema comprising the timestamp, its integer second and nanosecond-remainder components, the price, and the size. For the two exchange-traded assets the price and size are taken directly from the trade record. For foreign exchange, where no trade price is present, the mid-price is used instead.

2.2.2 Event Filtering

Observations that did not represent genuine events were removed before constructing the duration series. This is important because removing observations after durations have already been calculated would leave the surrounding durations unchanged and break the sequence of arrivals.

For GBP/USD, the data are quote observations rather than trades. Some records change quoted depth without changing either the bid or ask price. These were treated as non-events and removed. An observation was retained only when at least one quoted price changed. The first observation was retained because it has no previous quote for comparison. Repeated quotes with no price change could otherwise create artificial short durations. The Sunday market reopening was also excluded so that the weekend closure was not treated as part of the continuous duration process.

For NVDA, only observations from the regular US trading session were retained, excluding pre-market and post-market activity. The session was restricted to 13:30–20:00 UTC.

No filtering was applied to BTCUSDT because the series contains executed trades and operates continuously.

2.2.3 Consolidation of Simultaneous Events and Segmentation

Observations sharing the same timestamp are combined into a single event. The consolidated price is the volume-weighted average of its constituents and the consolidated size is their sum:

$$p = \frac{\sum_j p_j q_j}{\sum_j q_j}, \quad q = \sum_j q_j. \quad (2.2)$$

where the sums run over the records sharing a timestamp, p_j and q_j denote the price and size of the j th such record, p is the volume-weighted average price of the consolidated event, and q its total size. This is done because records with the same timestamp produce a zero duration,

which the continuous likelihood cannot accommodate. A zero duration means that no positive interval can be measured between the records at the available timestamp resolution.

The principle that simultaneous transactions constitute a single arrival is due to Engle and Russell [7]. The proportion of records requiring consolidation differs substantially across the three assets: none for foreign exchange, 49.51 per cent for the cryptocurrency series, and 48.50 per cent for the equity series. A non-price-revision in the FX quote feed is a non-economic observation and is therefore deleted. In contrast, observations with the same timestamp but different information are consolidated because they occur at the same observable instant.

Following consolidation, the events were divided into appropriate sessions before duration calculation. This was done to prevent duration from being calculated across a market boundary. The first observation in each session has no preceding observation within that session and therefore does not have a duration. For NVDA, each trading day forms a separate session because the regular market closes overnight. For GBP/USD, the implementation also uses calendar-day segmentation. This additionally prevents the weekly Friday-to-Sunday market closure from generating an artificial duration. For the GBP/USD, the raw data showed a reduction in activity around the daily Forex rollover period. The interval from 21:00 to 21:40 UTC was excluded, which was selected from the observed intra-day duration pattern. BTCUSDT was treated as a single continuous session because the market operates 24 hours a day, seven days a week.

2.2.4 Duration Construction

Durations are computed as the interval between consecutive events within a trading session, using Equation (1.1). The first event of each session is discarded, as the interval separating it from the final event of the preceding session measures the length of a closure rather than a waiting time between arrivals.

A session is a continuous period of trading containing no closure. For the cryptocurrency series the entire sample constitutes a single session. For the equity series each retained daily window forms one session. For foreign exchange the market trades continuously from Sunday evening to Friday evening, and the sample is segmented accordingly.

Care is taken throughout to ensure that no zero or negative durations enter the series, and that the first observation of each session is never used to generate a duration.

2.3 Sample Characteristics and Train-test Split

Table 2.1 summarises the three datasets. The three assets differ substantially in trading calendar and in the scale of their durations: GBP/USD trades on a five-day week, BTCUSDT continuously, and NVDA within a bounded daily session, giving 13, 19, and 13 trading days respectively over the sample period. The proportion of durations below one millisecond rises sharply with timestamp resolution, from zero for GBP/USD to 15.65% for BTCUSDT and 49.27% for NVDA. The duration levels are not directly comparable across the three assets because

GBP/USD records quote revisions, whereas BTCUSDT and NVDA record trades.

Table 2.1: Summary of the three datasets

	GBP/USD	BTCUSDT	NVDA
Schedule	24/5	24/7	13:30 to 20:00 / 5
Trading days	13	19	13
Raw transactions	1,247,646	6,110,740	2,555,159
Min. duration (s)	0.0500	1.0×10^{-6}	1.5×10^{-7}
Max. duration (s)	250.84	12.96	9.95
Mean duration (s)	0.8574	0.2686	0.1191
Median duration (s)	0.2040	0.0803	0.0011
% < 1 ms	0.000	15.647	49.268
% < 10 ms	0.000	30.694	62.976
% < 100 ms	9.549	53.164	78.157
% < 1 s	79.672	93.439	97.306

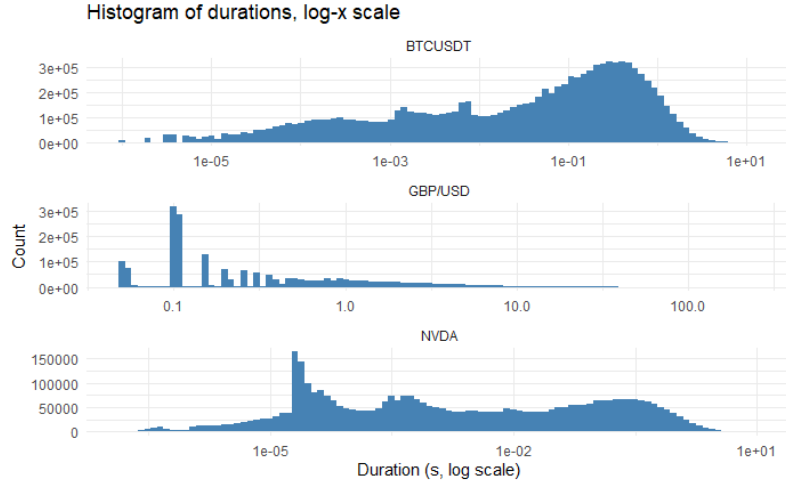


Figure 2.1: Distribution of raw durations for the three assets, logarithmic horizontal scale

Figure 2.1 shows the same durations on a logarithmic horizontal scale, to see the whole range at once, since the durations span roughly seven orders of magnitude. GBP/USD sits in a narrow band with a clear peak near 0.1 seconds, whereas BTCUSDT and NVDA spread across the entire range and NVDA carries a sharp spike close to 10^{-5} seconds, which is the sub-millisecond mass already reported in Table 2.1.

The sample is divided chronologically after 14 calendar days to form the training set. The following five days are retained as the out-of-sample test period. The division is chronological rather than random because durations are autocorrelated. A random split could place observations from the same burst of activity in both samples and weaken the separation between model

fitting and evaluation.

2.4 Diurnal Adjustment

Trading activity changes during the day. During busy periods, events occur more frequently and durations are shorter. During quieter periods, durations are longer. If this regular pattern is left in the data, an ACD model may treat it as duration clustering. The time-of-day pattern is therefore removed before the duration models are fitted.

Let x_i denote the raw duration between events $i - 1$ and i , as defined earlier, and let t_i denote the time of day of event i , measured in minutes from midnight UTC. The symbol t_i remains the full timestamp of event i . The adjusted duration is

$$\tilde{x}_i = \frac{x_i}{\hat{\phi}(t_i)}, \quad (2.3)$$

where $\tilde{x}_i$ is the adjusted duration and $\hat{\phi}(t_i)$ is the estimated diurnal factor at time t_i . A larger value of $\hat{\phi}(t_i)$ represents a time when durations are normally longer, while a smaller value represents a time when durations are normally shorter. Dividing by this factor removes the regular daily pattern while keeping all durations positive. The factor is estimated from the training data only.

Following Engle and Russell [7], the training data are first divided into time-of-day intervals. For the selected method, each interval is five minutes long. The mean duration in interval b is

$$m_b = \frac{1}{n_b} \sum_{i: t_i \in b} x_i, \quad (2.4)$$

where b identifies a five-minute interval, n_b is the number of observations in that interval, and m_b is their mean duration.

The interval means can be noisy, so a natural cubic spline is fitted to obtain a smooth time-of-day curve. The knots are placed every ten minutes. The spline coefficients are estimated by weighted least squares:

$$\hat{\beta} = \arg \min_{\beta} \sum_b n_b \left[m_b - \beta_0 - \sum_{k=1}^K \beta_k N_k(t_b) \right]^2, \quad (2.5)$$

where $\beta = (\beta_0, \beta_1, \dots, \beta_K)$ is the set of spline coefficients, $\hat{\beta}$ contains their estimated values, t_b is the time represented by interval b , and $N_k(t_b)$ is the k th natural cubic spline basis function evaluated at that time. The value K is the number of spline terms. The weight n_b gives more influence to intervals containing more observations because their mean durations are estimated more reliably.

The fitted diurnal factor is

$$\hat{\phi}(t) = \hat{\beta}_0 + \sum_{k=1}^K \hat{\beta}_k N_k(t), \quad (2.6)$$

where t is any time of day and $\hat{\beta}_0, \dots, \hat{\beta}_K$ are the estimated spline coefficients. The curve is evaluated on a common five-minute grid. Outside the time range observed in the training data, the factor is held at the nearest boundary value. This mainly concerns NVDA because it trades only during its exchange session.

Two versions of the procedure are compared. Method 1 uses 30-minute intervals with knots every 60 minutes, following the setting used by Engle and Russell [7]. Method 2 uses five-minute intervals with knots every ten minutes. Both methods keep the same ratio between interval width and knot spacing. The comparison therefore changes the time resolution without changing the relative spacing of the knots.

The root mean square error is

$$\text{RMSE} = \sqrt{\frac{1}{J} \sum_{j=1}^J [\hat{\phi}_r(t_j) - \bar{x}(t_j)]^2}, \quad (2.7)$$

where t_j is the j th five-minute grid point, J is the number of grid points included in the comparison, $\hat{\phi}_r(t_j)$ is the fitted factor for method r , and $\bar{x}(t_j)$ is the raw mean duration at the same time. Only grid points within the observed training range are used. A smaller RMSE means that the fitted curve is closer to the observed time-of-day profile.

A paired t -test is also used because the two methods are evaluated at the same grid points.

The null hypothesis is that the mean difference is zero.

Table 2.2: Comparison of the two diurnal-adjustment methods

Asset	Mean adjusted duration		Profile RMSE		Paired t -test	
	30-min	5-min	30-min	5-min	t	p
GBP/USD	0.9856	0.9924	0.4851	0.3718	3.52	4.95×10^{-4}
BTCUSDT	0.9402	0.9387	0.0413	0.0260	7.36	1.97×10^{-12}
NVDA	0.8596	0.9027	0.0281	0.0088	6.57	6.82×10^{-9}

Table 2.2 shows that the five-minute method has the lower RMSE for all three assets, and the paired t -tests give p -values below 0.05 with positive statistics, so the 30-minute method has the larger error. Five-minute intervals with ten-minute knots are therefore used throughout. Because errors at nearby times may be related, the t -test is treated as supporting evidence alongside the RMSE rather than as the sole basis for the choice.

Figure 2.2 compares the raw and adjusted profiles. The raw profiles change across the day while the adjusted ones are much flatter, so most of the time-of-day pattern has been removed. All duration models from this point onwards are fitted to $\tilde{x}_i$ rather than to the raw duration x_i .

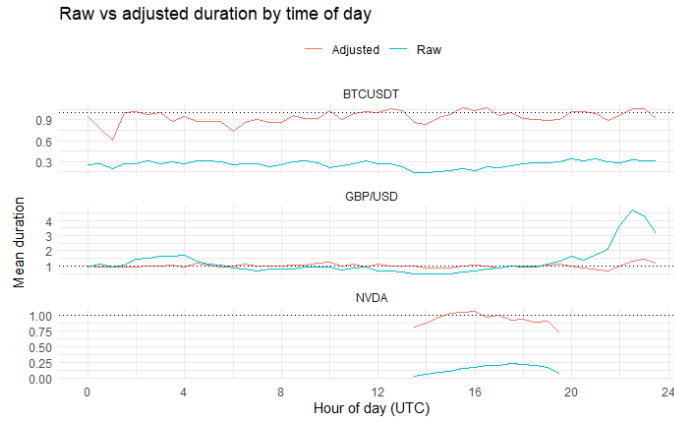


Figure 2.2: Mean duration by time of day before and after diurnal adjustment

Table 2.3: Dependence before and after diurnal adjustment

Asset	$\hat{\rho}_1$ before	$\hat{\rho}_1$ after	$Q(15)$ before	$Q(15)$ after
GBP/USD	0.1927	0.1436	687,453.0	339,408.3
BTCUSDT	0.2774	0.2730	1,219,484.2	1,164,133.4
NVDA	0.1653	0.1369	340,490.3	157,516.4

Table 2.3 shows that the lag-one autocorrelation $\hat{\rho}_1$ and the Ljung–Box statistic $Q(15)$ both fall after adjustment. $Q(15)$ drops by 50.6% for GBP/USD and 53.7% for NVDA but only 4.5% for BTCUSDT, which fits a market that trades continuously with no daily open or close.

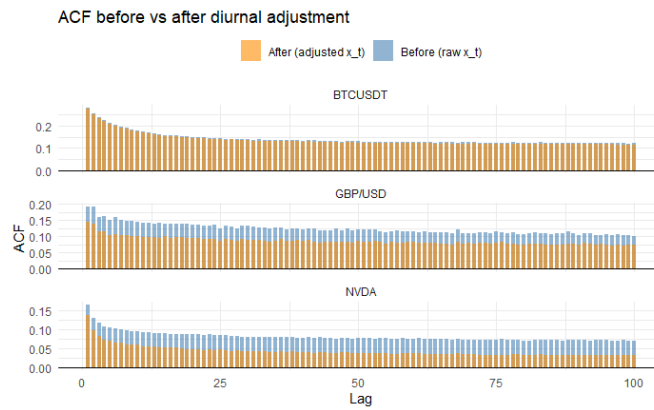


Figure 2.3: Autocorrelation functions before and after diurnal adjustment

Figure 2.3 shows the same comparison across all lags. The autocorrelation is lower after adjustment but stays clearly positive, and hence the remaining dependence is explained by the models.

Chapter 3

Model Specifications and Results

This chapter presents the models fitted to the duration series and their results. It begins with the homogeneous Poisson process and then considers linear and logarithmic ACD models with exponential and Weibull innovations.

3.1 The Homogeneous Poisson Process

3.1.1 Methodology

The simplest possible description of event arrivals is the homogeneous Poisson process, under which the conditional intensity, the instantaneous probability of an event in the next moment, is constant. Events arrive at a steady average rate, indifferent to the time of day and to whatever has happened before. Under this assumption the durations between events are independent and identically distributed exponential variables,

$$\tilde{x}_i \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda), \quad f(\tilde{x}) = \lambda e^{-\lambda \tilde{x}}, \quad \tilde{x} > 0, \quad (3.1)$$

where $\lambda > 0$ is the arrival rate, $1/\lambda$ is the expected duration, and $f(\tilde{x})$ is the exponential density. For an exponential distribution, the mean and standard deviation are equal, so the coefficient of variation

$$\text{CV} = \frac{\text{sd}(\tilde{x})}{\mathbb{E}[\tilde{x}]} \quad (3.2)$$

must equal exactly one.

Under the exponential null hypothesis, $\text{CV} = 1$, so $\text{CV}^2 - 1$ should equal zero. Whether the sample value differs significantly from this is tested with the delta method, which gives the approximate sampling distribution of $\text{CV}^2 - 1$ when CV is estimated from a finite sample and yields the asymptotically standard normal statistic

$$Z = \sqrt{\frac{n}{4}} (\text{CV}^2 - 1) \stackrel{\text{asy.}}{\sim} N(0, 1), \quad (3.3)$$

where n is the number of observed durations. At the five per cent significance level the null hypothesis is rejected when $|Z| > 1.96$.

The very large sample sizes used in this study make the test highly sensitive: even a small departure from $CV = 1$ can produce a significant Z , so statistical significance need not imply the departure is practically important. For this reason the magnitude of the departure, $CV^2 - 1$, is reported alongside Z , which together distinguish a statistically detectable deviation from one that is meaningfully large in practice.

3.1.2 Results

The homogeneous Poisson process is fitted to the diurnally adjusted durations, $\tilde{x}_i$, rather than the raw series. The rate is estimated by $\hat{\lambda} = 1/\bar{\tilde{x}}$, the reciprocal of the sample mean of the adjusted durations, computed over the complete training-period series for each asset.

Table 3.1 reports the fitted rate and the mean, median of the adjusted durations. The mean of the adjusted series is close to one for every asset by construction of the diurnal factor. The coefficient of variation takes values of 1.9827, 1.6355 and 2.4317 for GBP/USD, BTCUSDT and NVDA respectively, each substantially above the value of exactly one the model requires.

Table 3.1: Homogeneous Poisson estimates on adjusted durations

Asset	n	$\hat{\lambda}$	Mean	Median	CV
GBP/USD	906,431	1.0035	0.9964	0.2518	1.9827
BTCUSDT	4,535,411	1.0052	0.9948	0.3225	1.6355
NVDA	1,856,576	1.0391	0.9622	0.0107	2.4317

Table 3.2 reports the formal test of Equation (3.3). The excess-dispersion statistic exceeds one thousand for every asset, so the exponential assumption is rejected at any conventional significance level. The sample sizes are very large, so the size of the departure is more informative than the p -value alone. The values of $CV^2 - 1$ are 2.931 for GBP/USD, 1.675 for BTCUSDT, and 4.913 for NVDA. These are large departures from the value of zero required by the model.

Table 3.2: Excess-dispersion test, homogeneous Poisson

Asset	CV	$CV^2 - 1$	Z
GBP/USD	1.9827	2.931	1,395.4
BTCUSDT	1.6355	1.675	1,783.5
NVDA	2.4317	4.913	3,347.4

The results show that all three assets have substantial excess dispersion, with CV well above the exponential benchmark of 1. The strongest departure is observed for NVDA, followed by

GBP/USD and BTCUSDT. Overall, this shows that the transaction durations are clearly more variable than would be expected under a homogeneous Poisson process, suggesting that a model allowing for duration dependence and clustering, such as an ACD model, is more appropriate.

3.1.3 Conclusion

The homogeneous Poisson process is rejected for all three assets. This conclusion does not rest on statistical significance alone. The coefficients of variation are 1.9827 for GBP/USD, 1.6355 for BTCUSDT, and 2.4317 for NVDA, all well above the exponential benchmark of one, as shown in Table 3.2.

3.2 The Inhomogeneous Process

The homogeneous Poisson process was rejected because it assumes a constant arrival rate, and trading intensity is plainly not constant. The natural next step is to let the rate vary over time, replacing the fixed λ with $\lambda(t)$, giving what is called an *inhomogeneous* process. This describes only that the rate varies with time, not how or why, so it is a general category rather than a specification that can be estimated directly.

Part of that variation is simply the clock: there is a genuine daily rhythm, and the diurnal adjustment captures it. But dependence survives the adjustment, so the rate at any moment depends not only on the time of day but on what has just happened, a burst of rapid trading tends to be followed by more of the same, and a lull by more of a lull, regardless of the hour.

The Autoregressive Conditional Duration model supplies exactly this: a specific inhomogeneous process in which the rate is written $\lambda(t) = 1/\psi(t)$, with the conditional mean $\psi(t)$ allowed to depend on past durations through an autoregressive recursion, rather than on clock time alone. It is therefore treated here as the estimable specification of the inhomogeneous case. Once a model of this kind is fitted, the Ljung–Box statistic applied to its residuals becomes a genuine goodness-of-fit test, measuring whether the fitted recursion has absorbed the history-dependence that the diurnal adjustment alone could not. The ACD framework and its estimation are developed in Section 3.3.

3.3 ACD Model

3.3.1 Introduction

The Autoregressive Conditional Duration (ACD) model was introduced by Engle and Russell [7] to describe the clustering seen in the irregularly spaced times between high-frequency financial transactions, where busy periods tend to be followed by busy periods and quiet periods by quiet periods. The idea is borrowed directly from the Generalised Autoregressive Conditional Heteroskedasticity (GARCH) model of volatility: just as GARCH lets today’s variance depend

on recent variances and shocks, the ACD model lets the expected time between events depend on recently observed durations. In effect, the logic that GARCH applies to volatility clustering is transferred to the clustering of waiting times between trades.

3.3.2 Methodology

The Standard ACD Model

As established in Section 1.2, the raw duration between two consecutive events is $x_i = t_i - t_{i-1}$, where t_i is the time stamp of the i -th event. Throughout the modelling that follows we work with the diurnally adjusted duration $\tilde{x}_i$, and we write $\tilde{x}_{i-1}, \tilde{x}_{i-2}, \dots, \tilde{x}_1$ for the history of the process, that is, every duration observed up to and including observation $i - 1$.

The central quantity in the model is the conditional expected duration, ψ_i , that is, the duration we expect to observe next given everything seen so far:

$$\psi_i \equiv \mathbb{E}(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1). \quad (3.4)$$

Because ψ_i is a function of the past durations alone, it is known one step ahead: once the history has been observed, ψ_i is simply a fixed number.

The ACD model then writes each duration as a multiplicative error model, splitting it into its predictable scale ψ_i and an unpredictable innovation:

$$\tilde{x}_i = \psi_i \varepsilon_i, \quad \text{equivalently} \quad \varepsilon_i = \frac{\tilde{x}_i}{\psi_i}. \quad (3.5)$$

Here ε_i is the standardised duration, and it carries the following two properties:

- ε_i is the unpredictable error, assumed independent and identically distributed (i.i.d.);
- $\mathbb{E}[\varepsilon_i] = 1$. This unit-mean condition is what makes ψ_i the conditional mean of the duration, since taking expectations of Equation (3.5) gives $\mathbb{E}[\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1] = \psi_i \mathbb{E}[\varepsilon_i] = \psi_i$.

To reproduce the observed clustering, ψ_i is modelled as a linear function of the p most recent durations and the q most recent conditional means. This gives the ACD(p, q) specification:

$$\psi_i = \omega + \sum_{j=1}^p \alpha_j \tilde{x}_{i-j} + \sum_{k=1}^q \beta_k \psi_{i-k}, \quad (3.6)$$

where:

- $\omega > 0$ is the constant (baseline) term;
- $\alpha_j \geq 0$, for $j = 1, \dots, p$, measures how strongly the conditional mean reacts to the most recent durations;
- $\beta_k \geq 0$, for $k = 1, \dots, q$, measures how much of the previous conditional mean is carried forward;

- the sum $\sum_{j=1}^p \alpha_j + \sum_{k=1}^q \beta_k$ is the *persistence* of the process, which governs how long a shock to trading intensity takes to fade.

For compactness, the parameters to be estimated are collected in the vector

$$\boldsymbol{\theta} = (\omega, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)'. \quad (3.7)$$

Taking unconditional expectations of Equation (3.6), and using the fact that $\mathbb{E}[\tilde{x}_{i-j}] = \mathbb{E}[\psi_{i-k}] = \mu$ under stationarity, yields the unconditional mean duration:

$$\mathbb{E}[\tilde{x}_i] = \mu = \frac{\omega}{1 - \left(\sum_{j=1}^p \alpha_j + \sum_{k=1}^q \beta_k \right)}. \quad (3.8)$$

Whether this mean is well defined depends entirely on the persistence. The three possible cases are:

- Persistence below one, $\sum_{j=1}^p \alpha_j + \sum_{k=1}^q \beta_k < 1$: the process is stationary and the unconditional mean in Equation (3.8) is finite and positive. The market then has a stable long-run average pace to which durations revert.
- Persistence equal to one, $\sum_{j=1}^p \alpha_j + \sum_{k=1}^q \beta_k = 1$: the denominator collapses to zero and the unconditional mean diverges to infinity. A shock to the trading pace never dies out, so the process has, in effect, an infinite memory.
- Persistence above one, $\sum_{j=1}^p \alpha_j + \sum_{k=1}^q \beta_k > 1$: the denominator turns negative and the implied mean becomes negative. Since a negative duration is physically impossible, this signals that the model has broken down and is amplifying shocks explosively rather than absorbing them.

The restrictions $\omega > 0$, $\alpha_j \geq 0$, $\beta_k \geq 0$, and persistence below one are common sufficient conditions for a positive, stationary linear ACD model. They describe the standard theoretical case. The numerical estimation used for the reported results did not impose the coefficient and persistence restrictions. The fitted coefficients and persistence are therefore checked and discussed after each parameter table.

With the conditional mean specified, the likelihood can now be constructed. Because ψ_i depends only on information dated $i - 1$ and earlier, it is fixed once the history $\tilde{x}_{i-1}, \dots, \tilde{x}_1$ is conditioned upon. The standardising step in Equation (3.5) is then a change of scale by that fixed quantity, with Jacobian $|d\epsilon_i/d\tilde{x}_i| = 1/\psi_i$. The conditional density of the duration therefore follows directly from the density of the standardised duration,

$$g(\tilde{x}_i | \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}) = \frac{1}{\psi_i} g\left(\frac{\tilde{x}_i}{\psi_i} \middle| \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}\right). \quad (3.9)$$

Taken together, Equations (3.6) and (3.9) reduce the whole modelling problem to a single outstanding choice: the distribution of the standardised duration ε_i . It is precisely this choice that distinguishes the members of the ACD family from one another, and the two specifications used in this study are developed in Sections 3.4 and 3.5.

3.3.3 Goodness of fit and Residual Diagnostic measures

Once a specification has been estimated, two distinct questions are asked of the fit to understand how well the model is fitting the underlying data.

The first is whether the conditional mean has absorbed the dependence structure in the durations, whether the clustering that motivated moving beyond the Poisson framework has genuinely been captured. This is assessed with the Ljung-Box statistic

$$Q(L) = n(n+2) \sum_{k=1}^L \frac{\hat{\rho}_k^2}{n-k}, \quad (3.10)$$

where n is the number of observed durations, L is the number of autocorrelation lags included in the test, and $\hat{\rho}_k$ is the sample autocorrelation of the standardised residuals at lag k . The statistic is calculated for the standardised residuals, ε_i , and separately for their squares, ε_i^2 . The squared residuals examine whether any remaining dependence in the conditional variance has been captured by the model.

At the large sample sizes used throughout this dissertation, the Ljung-Box statistic is highly sensitive to even very small residual autocorrelations. Consequently, a statistically significant $Q(L)$ does not necessarily imply that the remaining dependence is practically important. The magnitude of the residual autocorrelation should therefore be considered alongside the formal test statistic.

Model fit is also compared using Akaike's Information Criterion (AIC) and the Bayesian Information Criterion (BIC):

$$\text{AIC} = -2\hat{\ell} + 2k, \quad \text{BIC} = -2\hat{\ell} + k \ln n, \quad (3.11)$$

where $\hat{\ell}$ is the maximised log-likelihood, k is the number of estimated parameters, and n is the sample size. Lower AIC and BIC values indicate a better balance between fit and model complexity. Given the very large samples used in this dissertation, statistical significance in the Ljung-Box tests is interpreted alongside the magnitude of the residual autocorrelation.

The second question is whether the assumed distribution for ε_i is the correct shape. This is assessed on the mean and dispersion of the standardised residuals, which under any correctly specified ACD model must satisfy $\mathbb{E}[\hat{\varepsilon}] \approx 1$, and, where the exponential distribution is assumed, $\text{sd}(\hat{\varepsilon}) \approx 1$ as well. A formal test of the latter, following the same over-powering logic, is

$$Z = \sqrt{\frac{n}{4}} (\text{CV}(\hat{\varepsilon})^2 - 1), \quad (3.12)$$

reported alongside the magnitude of the departure, $CV^2 - 1$, for the same reason.

3.4 Exponential ACD Model

3.4.1 Methodology

The Exponential ACD (EACD) model is the simplest member of the ACD family. It assumes that the standardised duration ε_i follows a standard exponential distribution.

The multiplicative structure of Equation (3.5) writes each adjusted duration as $\tilde{x}_i = \psi_i \varepsilon_i$, where $\psi_i = \mathbb{E}[\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1]$ is the conditional expected duration given the history $\tilde{x}_{i-1}, \dots, \tilde{x}_1$ available through observation $i - 1$, and ε_i is the independent and identically distributed (i.i.d.) innovation that carries the unpredictable part of the duration. The EACD model takes this innovation to be exponential with unit rate,

$$\varepsilon_i \sim \text{Exp}(1), \quad (3.13)$$

so that its probability density function is

$$g(\varepsilon_i) = \exp(-\varepsilon_i), \quad \varepsilon_i \geq 0. \quad (3.14)$$

Because ε_i is i.i.d., this density does not depend on the past history $\tilde{x}_{i-1}, \dots, \tilde{x}_1$. Throughout, $g(\cdot)$ denotes the same density introduced in the general framework of Equation (3.9); its argument distinguishes the density of the standardised duration, $g(\varepsilon_i)$, from the conditional density of the duration itself, $g(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta})$.

A standard exponential has mean one, which is exactly the moment condition $\mathbb{E}[\varepsilon_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1] = 1$ imposed on every ACD model for identification.

The density of the duration itself now follows from the density of ε_i . Substituting the exponential innovation density (3.14) into the general change-of-scale result of Equation (3.9), and recalling that ψ_i depends only on the history $\tilde{x}_{i-1}, \dots, \tilde{x}_1$ and is therefore fixed once that history is conditioned upon (so that the transformation contributes the Jacobian $1/\psi_i$), the conditional density of the adjusted duration $\tilde{x}_i$ becomes

$$g(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}) = \frac{1}{\psi_i} g\left(\frac{\tilde{x}_i}{\psi_i}\right) = \frac{1}{\psi_i} \exp\left(-\frac{\tilde{x}_i}{\psi_i}\right), \quad \tilde{x}_i \geq 0. \quad (3.15)$$

Estimation proceeds by maximum likelihood. Writing $\boldsymbol{\theta} = (\omega, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)'$ for the parameter vector defined in Equation (3.7), the conditional likelihood over the N observations is the product of the one-step-ahead densities from Equation (3.15),

$$L(\boldsymbol{\theta}) = \prod_{i=1}^N g(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}) = \prod_{i=1}^N \frac{1}{\psi_i} \exp\left(-\frac{\tilde{x}_i}{\psi_i}\right). \quad (3.16)$$

Taking logarithms turns this product into a sum and gives the conditional log-likelihood of the

EACD model,

$$\begin{aligned}
\ell(\boldsymbol{\theta}) &= \ln L(\boldsymbol{\theta}) = \sum_{i=1}^N \ln g(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}) \\
&= \sum_{i=1}^N \left[-\ln(\psi_i) - \frac{\tilde{x}_i}{\psi_i} \right] \\
&= - \sum_{i=1}^N \left[\ln(\psi_i) + \frac{\tilde{x}_i}{\psi_i} \right].
\end{aligned} \tag{3.17}$$

The parameter estimates $\hat{\boldsymbol{\theta}}$ are the values that maximise Equation (3.17). Because no closed-form solution exists, the maximisation is carried out numerically. The reported estimation does not force every coefficient to be non-negative or force persistence below one. This allows the estimates to show when the fitted model lies outside the standard stationary case. The resulting coefficients and persistence are discussed below Table 3.3.

3.4.2 Results

From Table 3.3, for GBP/USD, persistence ranges from 0.9982 to 0.9999, so every fitted order remains below one. The EACD(2,2) model has persistence of 0.9999 and is preferred within the EACD family because it has the lowest residual statistics, AIC, and BIC as shown in Table 3.4.

Table 3.3: EACD parameter estimates for all fitted orders

Asset	Order	ω	α_1	α_2	β_1	β_2	Persistence
GBP/USD	(1,1)	0.00258	0.0415	–	0.9571	–	0.9985
	(1,2)	0.00322	0.0522	–	0.6804	0.2656	0.9982
	(2,1)	0.00213	0.0831	–0.0464	0.9622	–	0.9988
	(2,2)	0.00027	0.0828	–0.0765	1.7108	–0.7173	0.9999
BTCUSDT	(1,1)	0.01084	0.2835	–	0.7529	–	1.0364
	(1,2)	0.00974	0.3308	–	0.4323	0.2810	1.0441
	(2,1)	0.00175	0.4442	–0.2956	0.8731	–	1.0217
	(2,2)	0.00001	0.3884	–0.3735	1.5546	–0.5672	1.0022
NVDA	(1,1)	0.00630	0.1080	–	0.9021	–	1.0102
	(1,2)	0.00809	0.1535	–	0.4401	0.4218	1.0154
	(2,1)	0.00189	0.3047	–0.2290	0.9351	–	1.0108
	(2,2)	0.00011	0.2573	–0.2396	1.5576	–0.5724	1.0029

For BTCUSDT, persistence ranges from 1.0022 to 1.0441, while for NVDA it ranges from 1.0029 to 1.0154. Every fitted EACD order is therefore above one for these two assets. The EACD(2,2) model still gives the best in-sample diagnostics within the family, but its above-one

persistence prevents the usual stationary interpretation. This issue is examined in the subsection 3.4.3.

Some second-order specifications contain negative coefficients because explicit parameter constraints were not imposed during estimation. Nevertheless, non-negative coefficients are sufficient, but not necessary, to ensure positive conditional durations, provided that the estimated parameter combination remains within a region that preserves positivity [16]. The persistence of each fitted model was therefore assessed after modelling as the sum of the autoregressive and moving-average duration coefficients, rather than being restricted to lie below unity during estimation.

Table 3.4 provides the order-selection diagnostics, and Figure 3.1 shows the residual autocorrelation at lags 1 to 15. The EACD(2,2) model gives the lowest values of Q_{resid} , Q_{resid}^2 , AIC, and BIC for all three assets. Its Q_{resid}^2 values are 31.3 for GBP/USD, 0.2 for BTCUSDT, and 0.8 for NVDA. The drift in the residual autocorrelation is also visible in Figure 3.1.

Table 3.4: EACD order-selection based on goodness-of-fit and residual diagnostics.

Asset	Order	Q_{adj}	Q_{resid}	Q_{resid}^2	AIC	BIC
GBP/USD	(1,1)	109,543.6	1,427.9	128.3	1,566,749	1,566,784
	(1,2)		1,275.3	102.4	1,566,082	1,566,129
	(2,1)		1,066.5	98.1	1,564,306	1,564,352
	(2,2)		230.4	31.3	1,559,640	1,559,698
BTCUSDT	(1,1)	2,181,271.6	13,039.9	61.1	5,873,227	5,873,267
	(1,2)		11,971.0	31.7	5,825,126	5,825,179
	(2,1)		9,255.4	2.2	5,723,936	5,723,989
	(2,2)		2,256.3	0.2	5,493,834	5,493,900
NVDA	(1,1)	144,521.7	5,000.0	14.4	2,751,256	2,751,293
	(1,2)		3,501.8	10.2	2,725,814	2,725,863
	(2,1)		2,124.5	1.1	2,664,759	2,664,809
	(2,2)		599.7	0.8	2,608,290	2,608,352

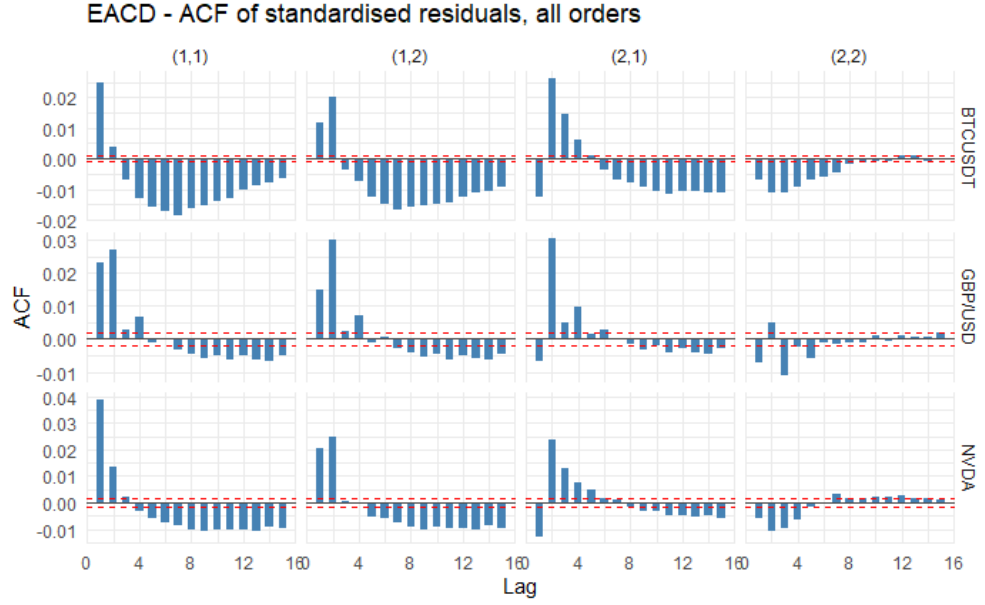


Figure 3.1: EACD: Sample autocorrelation function of the standardised residuals

3.4.3 Explanation of persistence above one for BTCUSDT and NVDA

Table 3.3 shows persistence above one for BTCUSDT and NVDA, but not for GBP/USD. Under the standard EACD interpretation, a value above one means that the fitted process does not have a finite positive unconditional mean. Two checks are therefore used to examine whether this result reflects the duration process itself or the recording of very closely spaced trades.

Merging near-simultaneous prints lowers persistence

As a first check, consecutive raw events were merged whenever the time between them was below a specified cut-off, after which the EACD(1,1) model was refitted. The procedure was repeated for cut-offs ranging from $1\ \mu\text{s}$ to 10 milliseconds.

Table 3.5: EACD(1,1) persistence after merging events closer than a given time cut-off

Merge cut-off	BTCUSDT n	BTCUSDT persist.	NVDA n	NVDA persist.
none (raw)	4,535,411	1.0367	1,856,523	1.0102
$1\ \mu\text{s}$	4,535,411	1.0367	1,836,514	1.0101
$10\ \mu\text{s}$	4,492,811	1.0336	1,726,173	1.0096
$100\ \mu\text{s}$	4,298,723	1.0237	1,280,960	1.0085
1 ms	3,835,868	1.0067	935,163	0.9982
10 ms	3,170,248	0.9943	687,930	0.9950

Table 3.5 shows that persistence decreases as the merging cut-off increases. For BTCUSDT,

it falls from 1.0367 in the raw data to 0.9943 at a 10 ms cut-off. For NVDA, it falls from 1.0102 to 0.9950. The $1\ \mu\text{s}$ BTCUSDT result is unchanged because the feed contains no events closer than one microsecond. Since the model and estimation procedure remain the same, this change is linked to the treatment of near-simultaneous trades. The result shows that very short durations have an important effect on the above-one persistence estimates. Once these records are merged, persistence moves below one for both assets.

Merging events also reduces the number of observations, so the fall in persistence could be caused by the smaller sample. To check this, a sensitivity analysis is done where shorter continuous sections of the original data are fitted without merging or removing durations. Table 3.6 shows that BTCUSDT persistence remains between 1.0297 and 1.0374, while NVDA remains between 1.0087 and 1.0136. These values stay close to the raw estimates and remain above one. The fall in Table 3.5 is therefore linked mainly to merging very short durations rather than to sample size alone.

Table 3.6: EACD(1,1) persistence when only sample size is reduced (no merging)

Asset	n used	Approx. % of training	Mean persist.	Range
BTCUSDT	4,000,000	88%	1.0357	1.0357 – 1.0357
BTCUSDT	2,000,000	44%	1.0341	1.0320 – 1.0353
BTCUSDT	1,000,000	22%	1.0330	1.0297 – 1.0374
NVDA	1,600,000	86%	1.0116	1.0115 – 1.0117
NVDA	800,000	43%	1.0111	1.0087 – 1.0123
NVDA	400,000	22%	1.0112	1.0097 – 1.0136

The second check, reported in Appendix B.3, rebuilds the events from information supplied by the data source instead of a cut-off chosen here. It gives persistence of 1.0085 for BTCUSDT and 1.0010 for NVDA, again much closer to one than the raw EACD(1,1) estimates of 1.0367 and 1.0102, so both the checks point the same way.

3.5 The Weibull ACD Model

3.5.1 Introduction

The Exponential ACD model provides a fundamental baseline for modelling transaction durations. Financial trade data, however, are frequently characterised by overdispersion. To capture a wider range of distribution shapes and accommodate this excess variance, Engle and Russell [7] proposed the Weibull Autoregressive Conditional Duration (WACD) model.

The WACD model achieves its flexibility by generalising the distributional assumption on the unpredictable error. Instead of the parameter-free exponential of the EACD model, the

standardised duration ε_i is assumed to follow a Weibull distribution with shape parameter $\gamma > 0$, which adds exactly one degree of freedom for controlling dispersion.

3.5.2 Methodology

The Weibull distribution with shape $\gamma > 0$ and scale $\lambda > 0$ has probability density function

$$g(\varepsilon_i; \gamma, \lambda) = \frac{\gamma}{\lambda} \left(\frac{\varepsilon_i}{\lambda} \right)^{\gamma-1} \exp \left(- \left(\frac{\varepsilon_i}{\lambda} \right)^\gamma \right), \quad \varepsilon_i \geq 0, \quad (3.18)$$

and mean $\lambda \Gamma(1 + 1/\gamma)$, where $\Gamma(\cdot)$ is the gamma function. The ACD framework requires the conditional mean of the duration to be captured entirely by ψ_i , that is $\mathbb{E}[\tilde{x}_i | \tilde{x}_{i-1}, \dots, \tilde{x}_1] = \psi_i$, which forces the unit-mean condition $\mathbb{E}[\varepsilon_i] = 1$. Imposing this fixes the scale parameter to

$$\lambda = \frac{1}{\Gamma(1 + 1/\gamma)}, \quad (3.19)$$

and λ is dependent on γ ; the shape parameter alone governs the distribution. Substituting Equation (3.19) into Equation (3.18) gives the density of the standardised duration used in the WACD model,

$$g(\varepsilon_i; \gamma) = \gamma \Gamma(1 + 1/\gamma) (\varepsilon_i \Gamma(1 + 1/\gamma))^{\gamma-1} \exp(-[\varepsilon_i \Gamma(1 + 1/\gamma)]^\gamma). \quad (3.20)$$

The Weibull has the exponential as a special case, when $\gamma = 1$ we have $\Gamma(1 + 1) = 1$, and Equation (3.20) collapses to $g(\varepsilon_i) = \exp(-\varepsilon_i)$, recovering the EACD specification. A fitted γ above or below one therefore measures how far the data departs from the exponential baseline.

The dynamics of the conditional mean ψ_i are unchanged: the WACD model uses the same autoregressive recursion as before, Equation (3.6), and alters only the shape of the noise around ψ_i , not the way past shocks are remembered through α_j and β_k . Applying the same change-of-scale argument used in Equation (3.9) and substituting into Equation (3.20) yields

$$g(\tilde{x}_i | \tilde{x}_{i-1}, \dots, \tilde{x}_1; \theta) = \frac{\gamma}{\tilde{x}_i} \left(\frac{\tilde{x}_i \Gamma(1 + 1/\gamma)}{\psi_i} \right)^\gamma \exp \left(- \left(\frac{\tilde{x}_i \Gamma(1 + 1/\gamma)}{\psi_i} \right)^\gamma \right). \quad (3.21)$$

Estimation is done by maximum likelihood. The parameter vector now extends the mean parameters of Equation (3.7) with the shape parameter,

$$\theta = (\omega, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q, \gamma)', \quad (3.22)$$

so that the conditional mean and the error shape are estimated jointly. The conditional likelihood over the N observations is the product of the one-step-ahead densities in Equation (3.21),

$$L(\theta) = \prod_{i=1}^N g(\tilde{x}_i | \tilde{x}_{i-1}, \dots, \tilde{x}_1; \theta), \quad (3.23)$$

and taking logarithms gives the conditional log-likelihood of the WACD model,

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^N \left[\ln\left(\frac{\gamma}{\tilde{x}_i}\right) + \gamma \ln\left(\frac{\tilde{x}_i \Gamma(1+1/\gamma)}{\psi_i}\right) - \left(\frac{\tilde{x}_i \Gamma(1+1/\gamma)}{\psi_i}\right)^\gamma \right]. \quad (3.24)$$

The estimates $\hat{\boldsymbol{\theta}}$ are obtained by maximising Equation (3.24) numerically. The only restriction imposed during estimation is the shape restriction $\gamma > 0$, which the Weibull density requires. As in the EACD case, the coefficients are not forced to be non-negative and persistence is not forced below one, so the fitted values are free to fall outside the standard stationary case and are checked afterwards against Table 3.7. Estimating the shape parameter jointly with the mean parameters lets the fitted error distribution adapt to the dispersion in the durations, which is the feature the exponential model cannot capture.

3.5.3 Results

Table 3.7 shows that the estimated Weibull shape parameter, $\hat{\gamma}$, is below one for every asset and order. It ranges from 0.7573 to 0.7593 for GBP/USD, from 0.5253 to 0.5324 for BTCUSDT, and from 0.2990 to 0.3006 for NVDA. The smaller estimates for BTCUSDT and NVDA reflect their stronger concentration of very short durations. For GBP/USD, persistence ranges from 0.9936 to 0.9993 and remains below one. For BTCUSDT, it ranges from 1.0202 to 1.2947, and for NVDA from 1.1384 to 1.5041. The WACD estimates for BTCUSDT and NVDA therefore remain outside the standard stationary case for every fitted order.

Table 3.7: WACD parameter estimates, all orders

Asset	Order	ω	α_1	α_2	β_1	β_2	$\hat{\gamma}$	Persist.
GBP/USD	(1,1)	0.00417	0.0391	–	0.9557	–	0.7573	0.9948
	(1,2)	0.00517	0.0489	–	0.6880	0.2567	0.7575	0.9936
	(2,1)	0.00352	0.0767	–0.0421	0.9609	–	0.7580	0.9956
	(2,2)	0.00052	0.0775	–0.0714	1.7030	–0.7099	0.7593	0.9993
BTCUSDT	(1,1)	0.00483	0.7061	–	0.5677	–	0.5253	1.2738
	(1,2)	0.00451	0.7666	–	0.4139	0.1142	0.5263	1.2947
	(2,1)	0.00247	0.8185	–0.2676	0.6589	–	0.5275	1.2098
	(2,2)	0.00003	0.7734	–0.7195	1.4201	–0.4538	0.5324	1.0202
NVDA	(1,1)	0.02057	0.7175	–	0.6763	–	0.2990	1.3938
	(1,2)	0.02327	0.9114	–	0.3339	0.2588	0.2994	1.5041
	(2,1)	0.00755	1.3329	–0.8645	0.8023	–	0.3002	1.2707
	(2,2)	0.00284	1.2807	–1.0425	1.1251	–0.2248	0.3006	1.1384

Table 3.8: WACD order-selection based on goodness-of-fit and residual diagnostics.

Asset	Order	Q_{resid}	Q_{resid}^2	Mean $\hat{\varepsilon}$	AIC	BIC
GBP/USD	(1,1)	1,426.8	136.4	1.0358	1,410,445	1,410,492
	(1,2)	1,269.3	109.9	1.0358	1,410,112	1,410,170
	(2,1)	1,056.2	104.7	1.0355	1,409,212	1,409,271
	(2,2)	188.2	30.9	1.0350	1,406,773	1,406,843
BTCUSDT	(1,1)	3,454.2	12.9	1.0934	1,575,752	1,575,806
	(1,2)	3,069.7	4.9	1.0895	1,569,081	1,569,148
	(2,1)	2,570.9	1.5	1.0829	1,562,997	1,563,064
	(2,2)	2,154.9	0.1	1.0544	1,514,279	1,514,359
NVDA	(1,1)	4,643.2	36.7	0.9243	-5,140,620	-5,140,570
	(1,2)	4,323.1	28.9	0.9235	-5,145,770	-5,145,708
	(2,1)	3,373.2	12.1	0.9078	-5,154,619	-5,154,556
	(2,2)	3,297.6	6.7	0.8975	-5,158,443	-5,158,369

For GBP/USD, WACD(2,2) gives the best results within the WACD family. Its Q_{resid} is 188.2, its Q_{resid}^2 is 30.9, and its AIC and BIC are 1,406,773 and 1,406,843. The shape estimate stays near 0.758 at every order and the mean standardised residual near 1.035, so the fitted error distribution behaves sensibly here.

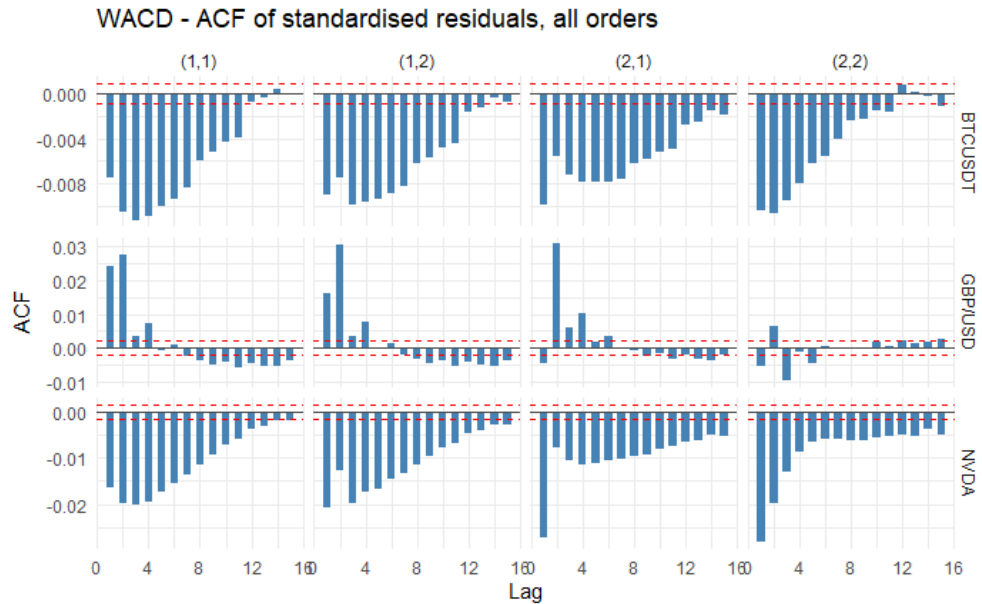


Figure 3.2: WACD: sample autocorrelation function of the standardised residuals

Table 3.8 and Figure 3.2 need to be read together, because they do not agree for every asset.

The table shows Q_{resid} falling with order throughout: from 1,426.8 to 188.2 for GBP/USD, from 3,454.2 to 2,154.9 for BTCUSDT and from 4,643.2 to 3,297.6 for NVDA. From the figure, for GBP/USD the fall is structural. The first two lags carry a clear positive spike, close to 0.03 in the first three orders, lag two is usually the taller of the pair, and at (2, 1) lag one is barely different from zero, and this collapses at (2, 2), leaving bars that are small and of mixed sign, so the extra lags repair an identifiable feature. Nothing of the kind happens for BTCUSDT and NVDA. Their residual autocorrelation is negative at every lag and at every order, decaying smoothly from roughly -0.009 and -0.02 at the shortest lags. That near-lag trough does not shrink as more lags are added, if anything it deepens slightly, and what actually falls is the tail beyond roughly lag eight, decaying to zero faster at higher order. A one-sided residual autocorrelation like this means the conditional mean is systematically misplaced rather than mis-timed, which a higher order cannot fix. So for these two assets the falling Q_{resid} reflects a fading tail rather than a genuinely corrected trough, not evidence of a better specification.

Table 3.7 points the same way. The shape estimate is small, from 0.5253 to 0.5324 for BTCUSDT and from 0.2990 to 0.3006 for NVDA against 0.7573 to 0.7593 for GBP/USD, and when $\gamma < 1$ the Weibull density rises without limit as the standardised duration approaches zero. The mean standardised residual drifts away from one, to between 1.0544 and 1.0934 for BTCUSDT and between 0.8975 and 0.9243 for NVDA, while GBP/USD stays between 1.0350 and 1.0358. The coefficients are unstable, with NVDA reaching $\hat{\alpha}_1 = 1.3329$ at (2, 1) and $\hat{\alpha}_2 = -1.0425$ at (2, 2), and persistence stays above one at every order for both assets. The NVDA AIC values are also large and negative, from $-5,140,620$ to $-5,158,443$, while every other model here gives positive values. AIC is not scale free, so a density diverging at zero, fitted to a series with 49.27 per cent of durations below one millisecond, drives it down without describing the data any better.

The WACD fits for BTCUSDT and NVDA are therefore driven by the mass of durations near zero rather than by the duration dependence, and they are not carried forward as candidate models for those two assets. GBP/USD shows none of these features, so its WACD fit is retained.

3.6 Logarithmic ACD (LACD) Model

3.6.1 Introduction

The standard ACD model of Engle and Russell [7] models the conditional duration directly and requires parameter restrictions to ensure positive durations. Bauwens and Giot [1] proposed the Log-ACD (LACD) model, which instead models the logarithm of the conditional duration. This provides an alternative way of modelling the duration dynamics, which is useful for BTCUSDT and NVDA, because they contain a large number of very short, sub-millisecond durations. In these two datasets, the EACD and WACD models produced persistence estimates above one, raising concerns about stationarity. Hence, LACD is used as an alternative logarithmic specification that may provide a useful representation of the duration dynamics. Two types called

LACD Type I and LACD Type 2 are used in this section.

3.6.2 Methodology

Rather than writing the recursion for the conditional expected duration ψ_i directly, the Log-ACD models write it for the natural logarithm of that quantity, $\ln \psi_i$. A duration can never be negative, so ψ_i is confined to the positive half-line, whereas $\ln \psi_i$ ranges over the whole real line. Whatever value the right-hand side of the recursion takes, exponentiating it back through $\psi_i = \exp(\ln \psi_i)$ always returns a positive number. This is what removes the sign restrictions that the linear ACD must impose by hand. Bauwens and Giot [1] gave two ways of feeding past information into the log recursion.

Log-ACD Type I (LACD1). The first specification, implemented in the ACDm package [2], drives the recursion with the logarithm of the past standardised durations:

$$\ln \psi_i = \omega + \sum_{j=1}^p \alpha_j \ln \varepsilon_{i-j} + \sum_{k=1}^q \beta_k \ln \psi_{i-k}, \quad (3.25)$$

where $\varepsilon_{i-j} = \tilde{x}_{i-j}/\psi_{i-j}$ is the standardised duration, ω is the constant, the α_j are the coefficients on the p past innovations, and the β_k are the coefficients on the q past log conditional means.

Log-ACD Type II (LACD2). The second specification, following the idea behind the EGARCH model of Nelson [12] and used by Lunde [10], brings the same past information in through the standardised duration itself rather than its logarithm:

$$\ln \psi_i = \omega + \sum_{j=1}^p \alpha_j \varepsilon_{i-j} + \sum_{k=1}^q \beta_k \ln \psi_{i-k}. \quad (3.26)$$

The two forms differ only in whether α_j acts on $\ln \varepsilon_{i-j}$ or on ε_{i-j} ; in both, the β_k terms act on the lagged log conditional mean $\ln \psi_{i-k}$.

Persistence in the Log-ACD Model

Persistence in these models is the autoregressive (AR) coefficient of the process once it is written in autoregressive moving-average (ARMA) form, since that coefficient governs how slowly a shock decays. Whether a given parameter belongs to the AR part depends on what it multiplies: a lagged quantity that carries the past conditional mean feeds the level forward and is autoregressive, whereas a lagged pure shock does not and is only moving-average. This helps in explaining why the linear and log models report persistence differently.

In the linear ACD, the recursion $\psi_i = \omega + \alpha \tilde{x}_{i-1} + \beta \psi_{i-1}$ has α acting on the duration $\tilde{x}_{i-1}$. This is not a shock: since $\tilde{x}_{i-1} = \psi_{i-1} \varepsilon_{i-1}$, its conditional mean is $\mathbb{E}[\tilde{x}_{i-1} | \tilde{x}_{i-2}, \dots, \tilde{x}_1] = \psi_{i-1}$, so it carries the past level. Rewriting the model in terms of ψ_i and the innovation $v_i = \tilde{x}_i - \psi_i$ (which

has mean zero) collapses it to an ARMA(1,1) in the duration with AR coefficient $(\alpha + \beta)$; both parameters transmit the level, so persistence is $\alpha + \beta$. The algebra is given in Section 3.3.2.

The Log-ACD is different because the standardised duration enters as a genuine shock. Writing $y_i \equiv \ln \psi_i$, the LACD1 recursion of Equation (3.25) is

$$y_i = \omega + \sum_{k=1}^q \beta_k y_{i-k} + \sum_{j=1}^p \alpha_j \ln \varepsilon_{i-j}, \quad (3.27)$$

with LACD2 identical but for $\ln \varepsilon_{i-j}$ replaced by ε_{i-j} . Here the two terms are already separated by role: $y_{i-k} = \ln \psi_{i-k}$ is the lagged conditional mean, carried forward, while $\varepsilon_{i-j} = \tilde{x}_{i-j} / \psi_{i-j}$ is an i.i.d. shock with $\mathbb{E}[\varepsilon_{i-j}] = 1$. Using the lag operator L ($L^k y_i = y_{i-k}$), Equation (3.27) is therefore already in ARMA form,

$$\left(1 - \sum_{k=1}^q \beta_k L^k\right) y_i = \omega + \sum_{j=1}^p \alpha_j \ln \varepsilon_{i-j}, \quad (3.28)$$

where the β_k are the AR coefficients and the α_j the moving-average coefficients on the shocks. No substitution is needed, because unlike $\tilde{x}_{i-1}$ in the linear case, ε_{i-j} contains no level to fold into the AR part. The process is stationary when the roots of

$$1 - \sum_{k=1}^q \beta_k z^k = 0 \quad (3.29)$$

lie outside the unit circle. When all β_k are non-negative, $\sum_k \beta_k < 1$ is a simple sufficient condition. Some fitted second-order models contain a negative $\hat{\beta}_2$, so their stationarity is checked from the characteristic roots rather than from the coefficient sum alone. These roots are reported in Table B.3. For comparison across orders, the reported persistence summary is

$$\text{Persistence}_{\text{LACD}} = \sum_{k=1}^q \beta_k. \quad (3.30)$$

The contrast is structural. In the linear ACD, α_j acts on past durations and contributes to persistence. In the Log-ACD, α_j acts on past innovations, so persistence is described by the β_k terms. Both measures describe how dependence carries forward, but they operate on different scales and are not directly comparable.

Log-Exponential ACD (Log-EACD)

Once the conditional mean has been defined through either logarithmic recursion, the model is completed by attaching an error distribution, such as the exponential or the Weibull. Take the baseline case in which the standardised shock follows a standard exponential distribution, which

preserves the unit-mean condition $\mathbb{E}[\varepsilon_i] = 1$:

$$\varepsilon_i \sim \text{Exp}(1). \quad (3.31)$$

The observed adjusted duration and the error are linked, as before, by the multiplicative structure $\tilde{x}_i = \psi_i \varepsilon_i$. Applying the same change-of-scale argument as Equation (3.9), the conditional density of the duration is identical in form to that of the standard Exponential ACD model,

$$g(\tilde{x}_i \mid \tilde{x}_{i-1}, \dots, \tilde{x}_1; \boldsymbol{\theta}) = \frac{1}{\psi_i} \exp\left(-\frac{\tilde{x}_i}{\psi_i}\right). \quad (3.32)$$

The whole difference lies inside ψ_i , which is now generated by the exponentiated Log-ACD recursion rather than the linear one.

Estimation is again by maximum likelihood, over the unrestricted parameter vector $\boldsymbol{\theta} = (\omega, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)'$. Taking the natural logarithm of the density in Equation (3.32) and summing across the N observations gives the conditional log-likelihood,

$$\ell(\boldsymbol{\theta}) = -\sum_{i=1}^N \left[\ln(\psi_i) + \frac{\tilde{x}_i}{\psi_i} \right], \quad (3.33)$$

where ψ_i is computed at each step from either the LACD1 or LACD2 recursion. The objective is the same as for the standard EACD in Equation (3.17); only the recursion generating ψ_i has changed.

3.6.3 Results

Table 3.9: LACD1 parameter estimates, all orders

Asset	Order	ω	α_1	α_2	β_1	β_2	Persist.
GBP/USD	(1,1)	0.04663	0.0519	–	1.0001	–	1.0001
	(1,2)	0.04616	0.0514	–	0.8510	0.1502	1.0013
	(2,1)	0.04074	0.0959	–0.0507	1.0017	–	1.0017
	(2,2)	0.00834	0.0996	–0.0903	1.7293	–0.7290	1.0003
BTCUSDT	(1,1)	0.19597	0.1589	–	0.8023	–	0.8023
	(1,2)	0.21867	0.1748	–	0.5870	0.2090	0.7960
	(2,1)	0.14616	0.1849	–0.0701	0.8756	–	0.8756
	(2,2)	0.00407	0.1685	–0.1656	1.7004	–0.7020	0.9984
NVDA	(1,1)	0.23080	0.0650	–	0.8584	–	0.8584
	(1,2)	0.23074	0.0660	–	0.7297	0.1289	0.8587
	(2,1)	0.13825	0.0934	–0.0555	0.9445	–	0.9445
	(2,2)	0.03478	0.0885	–0.0791	1.5266	–0.5372	0.9894

Table 3.9 reports the LACD1 estimates across all four candidate orders. Persistence is computed as $\sum \hat{\beta}_k$ alone, following the reasoning set out in Section 3.6.2.

For GBP/USD, persistence sits close to the unit boundary at every order, ranging from 1.0001 to 1.0017, a range so narrow that the choice of order makes almost no difference to this figure, mirroring the same order-stability already observed for EACD (Table 3.3).

For BTCUSDT, the LACD1(2,2) specification produces a persistence of 0.9984, made up of a large first-order coefficient ($\hat{\beta}_1 = 1.7004$) offset by a negative second-order coefficient ($\hat{\beta}_2 = -0.7020$). This points to a more involved dynamic structure in which the effect of past conditional durations is spread over more than one lag.

For NVDA, the log-scale persistence ranges from 0.8584 to 0.9894. The LACD1(1,1) model gives $\hat{\beta}_1 = 0.8584$, while the (1,2) and (2,1) specifications give persistence values of 0.8587 and 0.9445 respectively.

The persistence measure reported for LACD1 is defined on the logarithmic conditional-duration scale and is therefore not directly comparable with the $\alpha + \beta$ persistence measure of the linear EACD and WACD models. A value below unity in the LACD1 specification should not be interpreted as evidence that the underlying duration process is stationary relative to an EACD or WACD specification. The persistence values are used primarily to characterise the dynamics within the LACD1 model class, while model selection is based on the overall goodness-of-fit and residual diagnostics reported in Table 3.10.

Table 3.10: LACD1 order-selection based on goodness-of-fit and residual diagnostics

Asset	Order	Q_{resid}	Q^2_{resid}	AIC	BIC
GBP/USD	(1,1)	12,469.5	1,171.8	1,427,132	1,427,167
	(1,2)	14,553.6	1,564.8	1,587,166	1,587,212
	(2,1)	13,161.3	1,209.2	1,583,350	1,583,397
	(2,2)	11,272.3	1,022.1	1,603,353	1,603,411
BTCUSDT	(1,1)	60,757.4	459.0	5,795,254	5,795,294
	(1,2)	55,801.2	296.5	5,748,798	5,748,851
	(2,1)	56,922.9	220.7	5,716,134	5,716,187
	(2,2)	61,780.4	614.2	5,516,443	5,516,510
NVDA	(1,1)	32,355.1	2,427.0	2,997,735	2,997,772
	(1,2)	33,206.9	2,744.7	2,982,198	2,982,247
	(2,1)	32,181.2	3,332.3	2,932,423	2,932,472
	(2,2)	33,950.2	3,484.3	2,929,597	2,929,659

Table 3.10 and Figure 3.3 show that LACD1 absorbs the dependence poorly. The first lag or two sit well above the rest, roughly 0.04 to 0.08 depending on the asset, and beyond that the remaining lags settle into a roughly constant 0.02 to 0.04 band. Neither part of this pattern

decays or improves as more lags are added, so Q_{resid} barely moves in consequence. GBP/USD and NVDA are selected at (2,2) and (1,1) respectively on residual diagnostics rather than AIC, which falls monotonically with added lags and is read here as over-fitting at this sample size; BTCUSDT is selected at (2,1).



Figure 3.3: LACD1: sample autocorrelation function of the standardised residuals

Table 3.11: LACD2 parameter estimates, all orders

Asset	Order	ω	α_1	α_2	β_1	β_2	Persist.
GBP/USD	(1,1)	-0.03688	0.0361	–	0.9948	–	0.9948
	(1,2)	-0.04596	0.0450	–	0.7357	0.2579	0.9936
	(2,1)	-0.03435	0.0619	-0.0283	0.9955	–	0.9955
	(2,2)	-0.00423	0.0627	-0.0585	1.8290	-0.8294	0.9996
BTCUSDT	(1,1)	-0.14498	0.1337	–	0.9670	–	0.9670
	(1,2)	-0.17724	0.1638	–	0.6886	0.2725	0.9611
	(2,1)	-0.12556	0.1917	-0.0752	0.9740	–	0.9740
	(2,2)	-0.00334	0.1867	-0.1834	1.8719	-0.8721	0.9998
NVDA	(1,1)	-0.06832	0.0637	–	0.9826	–	0.9826
	(1,2)	-0.09262	0.0863	–	0.6010	0.3760	0.9770
	(2,1)	-0.06002	0.1182	-0.0620	0.9864	–	0.9864
	(2,2)	-0.00795	0.1087	-0.1012	1.7901	-0.7915	0.9987

Table 3.11 reports the LACD2 estimates across all four candidate orders. For GBP/USD,

persistence again sits close to the unit boundary at every order, from 0.9936 to 0.9996, staying almost flat regardless of order, the same pattern already seen for EACD.

For BTCUSDT, persistence rises from 0.9611 at (1,2) to 0.9998 at (2,2), approaching the boundary as order increases.

For NVDA, persistence rises from 0.9770 at (1,2) to 0.9987 at (2,2), the same upward trend. In both cases the reading is close to one but still below it.

Several of these second-order fits carry a negative $\hat{\beta}_2$, so their stationarity cannot be settled by the coefficient sum alone and is checked from the characteristic roots instead. Appendix B.4 reports those roots. All three LACD2(2,2) recursions are stationary on the log scale, though only just, with smallest roots of 1.0024 for GBP/USD, 1.0016 for BTCUSDT and 1.0069 for NVDA. The LACD1(2,2) fit selected for GBP/USD is not stationary: its smallest root is 0.9989, which lies inside the unit circle.

The Table 3.12 and Figure 3.4 show a consistent pattern where Q_{resid} , Q^2_{resid} , AIC and BIC all fall as order increases, and (2,2) is preferred for all three assets with no trade-off to weigh. In the figure the lag-one spike present at (1,1) moves to lag two at (1,2) and (2,1) and then largely disappears at (2,2). The contrast with Figure 3.3 is marked given that the two specifications share the same log-scale recursion and differ only in whether α_j multiplies $\ln \varepsilon_{i-j}$ or ε_{i-j} , which locates the difference in the specification rather than in estimation. The estimated ω is negative throughout, which the logarithmic form accommodates without difficulty since $\psi_i = \exp(\ln \psi_i)$ remains positive by construction.

Table 3.12: LACD2 order-selection based on goodness-of-fit and residual diagnostics

Asset	Order	Q_{resid}	Q^2_{resid}	AIC	BIC
GBP/USD	(1,1)	1,395.0	95.1	1,541,919	1,541,954
	(1,2)	1,236.1	85.5	1,541,225	1,541,272
	(2,1)	1,026.2	88.5	1,539,928	1,539,975
	(2,2)	210.9	72.0	1,535,231	1,535,289
BTCUSDT	(1,1)	15,290.6	24.4	5,973,385	5,973,425
	(1,2)	12,482.3	16.8	5,941,142	5,941,196
	(2,1)	11,774.7	8.7	5,914,165	5,914,218
	(2,2)	994.0	4.2	5,708,460	5,708,527
NVDA	(1,1)	5,209.4	16.8	2,715,525	2,715,562
	(1,2)	3,501.1	14.8	2,697,199	2,697,249
	(2,1)	2,051.7	10.4	2,676,107	2,676,157
	(2,2)	443.8	5.4	2,640,640	2,640,702

As with LACD1, the persistence figures reported here are on the logarithmic scale and are not directly comparable to the linear $\alpha + \beta$ persistence reported for EACD and WACD. The

finding that BTCUSDT and NVDA are integrated processes continues to rest on the EACD results alone; the LACD2 results are used here only to choose the best-fitting specification within the LACD2 family, on the basis of Q_{resid}^2 and Q_{resid} .

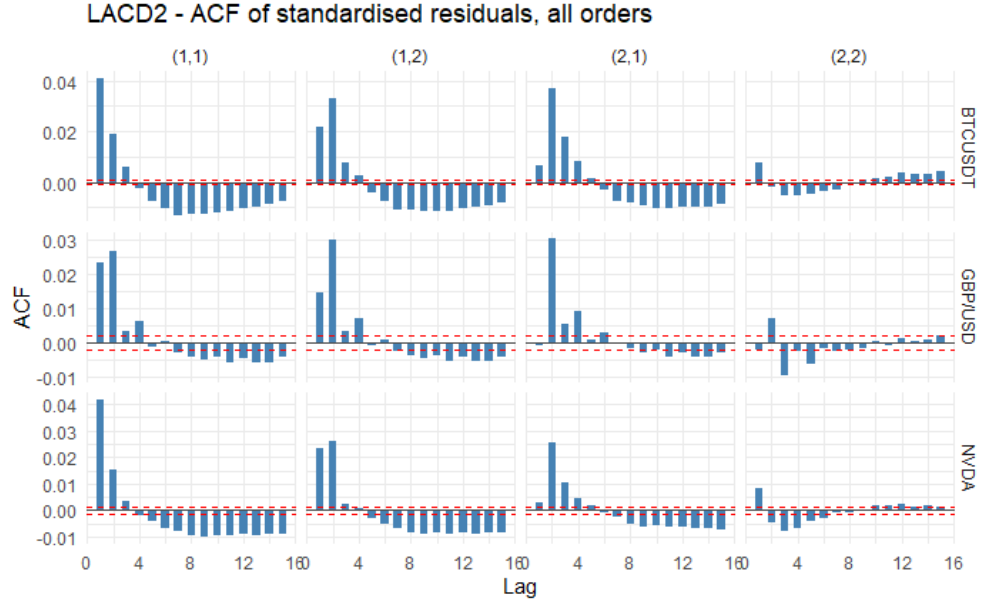


Figure 3.4: LACD2: sample autocorrelation function of the standardised residuals

3.7 Cross-model comparison

This section compares the best specification from each family, per asset, to arrive at one preferred model. Two rules carry over from earlier sections. WACD is excluded for BTCUSDT and NVDA for the reasons given in Section 3.5, and persistence is not used as a criterion across families, because the linear and logarithmic specifications measure it on different scales.

For GBP/USD, WACD(2,2) is lowest on all four measures at once: Q_{resid} of 188.2, Q_{resid}^2 of 30.9, and the lowest AIC and BIC. EACD(2,2) is close on the residual statistics, at 230.4 and 31.3, but well above on AIC and BIC. LACD2(2,2) matches on Q_{resid} at 210.9 but more than doubles Q_{resid}^2 at 72.0, and LACD1(2,2) trails throughout. WACD(2,2) is preferred with no trade-off to weigh, and its persistence of 0.9993, against 0.9999 for EACD(2,2), places the asset just inside the stationary region.

For BTCUSDT and NVDA the measures disagree, so the choice separates into two questions.

On fit alone EACD(2,2) wins. It gives the lowest Q_{resid}^2 of any specification tested, 0.2 and 0.8 against 4.2 and 5.4 for LACD2(2,2), and the lowest Q_{resid} , AIC and BIC for both assets.

Table 3.13: Best specification within each model family, compared per asset

Asset	Model	Order	Q_{resid}	Q_{resid}^2	AIC	BIC
GBP/USD	EACD	(2,2)	230.4	31.3	1,559,640	1,559,698
	WACD	(2,2)	188.2	30.9	1,406,773	1,406,843
	LACD1	(2,2)	11,272.3	1,022.1	1,603,353	1,603,411
	LACD2	(2,2)	210.9	72.0	1,535,231	1,535,289
BTCUSDT	EACD	(2,2)	2,256.3	0.2	5,493,834	5,493,900
	LACD1	(2,1)	56,922.9	220.7	5,716,134	5,716,187
	LACD2	(2,2)	994.0	4.2	5,708,460	5,708,527
NVDA	EACD	(2,2)	599.7	0.8	2,608,290	2,608,352
	LACD1	(1,1)	32,355.1	2,427.0	2,997,735	2,997,772
	LACD2	(2,2)	443.8	5.4	2,640,640	2,640,702

It cannot be used beyond one step ahead. Its persistence is 1.0022 and 1.0029, and by Section 3.3.2 a linear ACD process above one has no finite unconditional mean, so a freely simulated path drifts instead of settling. LACD2 has no such problem: its log-scale recursion is stationary for both assets, with smallest characteristic roots of 1.0016 and 1.0069 (Table B.3), and $\psi_i = \exp(\ln \psi_i)$ stays positive whatever the recursion returns.

LACD2(2,2) is therefore carried forward for both assets. Its logarithmic form suits these two series because BTCUSDT has 15.65% of its durations below one millisecond and NVDA 49.27%, while GBP/USD has none. The linear recursion must stay positive while working with durations close to zero, and Table 3.3 shows the cost: negative $\hat{\alpha}_2$ and $\hat{\beta}_2$, $\hat{\omega}$ collapsing towards zero, and persistence pushed above one. The log recursion acts on $\ln \psi_i$, which may take any real value, so a run of very short durations lowers it without threatening positivity. GBP/USD, the one asset with no sub-millisecond durations, is also the one where the linear model stays below one. The choice therefore rests on stability and on the handling of short durations, not on any claim that LACD2 explains the dependence better than EACD, which it does not.

Chapter 4

Simulation Study

LACD2(2,2) was preferred for BTCUSDT and NVDA in Section 3.7 partly because EACD(2,2), the better in-sample fit, cannot generate a series of its own. That argument is only worth making if the chosen models are put to work. Each is therefore simulated over the test period and compared with the observed durations, with the homogeneous Poisson process of Section 3.1 alongside as a baseline. Already rejected in sample, it sets a floor that a model with more parameters ought to clear.

4.1 Method

For GBP/USD the fitted WACD(2,2) recursion is run forward,

$$\psi_i = \hat{\omega} + \sum_{j=1}^2 \hat{\alpha}_j \tilde{x}_{i-j} + \sum_{k=1}^2 \hat{\beta}_k \psi_{i-k}, \quad \tilde{x}_i = \psi_i \varepsilon_i, \quad (4.1)$$

with $\varepsilon_i \sim \text{Weibull}(\hat{\gamma}, 1/\Gamma(1 + 1/\hat{\gamma}))$. For BTCUSDT and NVDA the LACD2(2,2) recursion is used,

$$\ln \psi_i = \hat{\omega} + \sum_{j=1}^2 \hat{\alpha}_j \varepsilon_{i-j} + \sum_{k=1}^2 \hat{\beta}_k \ln \psi_{i-k}, \quad \tilde{x}_i = \psi_i \varepsilon_i, \quad (4.2)$$

with $\varepsilon_i \sim \text{Exp}(1)$. The baseline draws $\tilde{x}_i \sim \text{Exp}(\hat{\lambda})$ independently, using the rate from Table 3.1.

Both recursions are seeded from the end of the training period, taking ψ_0 and ψ_{-1} from the last fitted conditional means and $\tilde{x}_0$ and $\tilde{x}_{-1}$ from the last observed training durations. From the first step onwards it feeds only on its own output.

Each model is simulated 200 times, with every path matching that asset's own test length: 341,215 durations for GBP/USD, 1,575,329 for BTCUSDT and 698,583 for NVDA. The recursions run on the adjusted scale, and each path is returned to raw seconds using the diurnal factor of the corresponding test observation,

$$\tilde{x}_i^{\text{raw}} = \hat{\phi}(t_i) \tilde{x}_i, \quad (4.3)$$

with $\hat{\phi}$ as defined in Equation (2.6).

4.2 Distance measures

The 99 percentiles of each path are averaged across the 200 paths to give a mean percentile curve $\bar{q}_u$, compared with the observed curve q_u^{obs} :

$$\text{quantile MAE} = \frac{1}{99} \sum_{u=1}^{99} |\bar{q}_u - q_u^{\text{obs}}|, \quad \text{quantile RMSE} = \sqrt{\frac{1}{99} \sum_{u=1}^{99} (\bar{q}_u - q_u^{\text{obs}})^2}. \quad (4.4)$$

The autocorrelation at lags 1 to 15 is computed separately for each path and then averaged, giving $\bar{\rho}_\ell$ against the observed ρ_ℓ^{obs} :

$$\text{ACF MAE} = \frac{1}{15} \sum_{\ell=1}^{15} |\bar{\rho}_\ell - \rho_\ell^{\text{obs}}|, \quad \text{ACF RMSE} = \sqrt{\frac{1}{15} \sum_{\ell=1}^{15} (\bar{\rho}_\ell - \rho_\ell^{\text{obs}})^2}. \quad (4.5)$$

Taking the autocorrelation per path before averaging matters: joining the paths end to end would create an artificial break at every join and depress the measured dependence.

Comparing simulated and observed durations index by index is not meaningful, since each of the 200 paths is an independent random draw and $\tilde{x}_i$ is not expected to match the observed series at the same position, only to share its statistical properties. A pointwise MAE or RMSE would therefore measure random-noise misalignment rather than model fit. The comparison instead targets the two properties an ACD specification claims to get right: the distribution of durations, via the quantile MAE/RMSE below, and the dependence structure, via the ACF MAE/RMSE that follows.

4.3 Results

Table 4.1: Distance between simulated and observed test durations, averaged over 200 paths

Asset	Model	Quantile MAE	Quantile RMSE	ACF MAE	ACF RMSE
BTCUSDT	Homogeneous Poisson	0.1032	0.1668	0.1356	0.1414
	LACD2(2,2)	0.0929	0.1623	0.0654	0.0718
GBP/USD	Homogeneous Poisson	0.3177	0.4147	0.0512	0.0541
	WACD(2,2)	0.1128	0.2046	0.0775	0.0778
NVDA	Homogeneous Poisson	0.0651	0.0981	0.0834	0.0859
	LACD2(2,2)	0.0577	0.1026	0.1672	0.1674

On quantile MAE the selected model beats the baseline for all three assets: 0.1032 to

0.0929 for BTCUSDT, 0.0651 to 0.0577 for NVDA, and 0.3177 to 0.1128 for GBP/USD, where WACD(2,2) more than halves the distance. The RMSE column is less flattering: for NVDA it moves the wrong way, 0.0981 to 0.1026, so LACD2 is closer on average but further away where it does worst. Squared error weights the extremes, and the extremes are where the sub-millisecond durations sit.

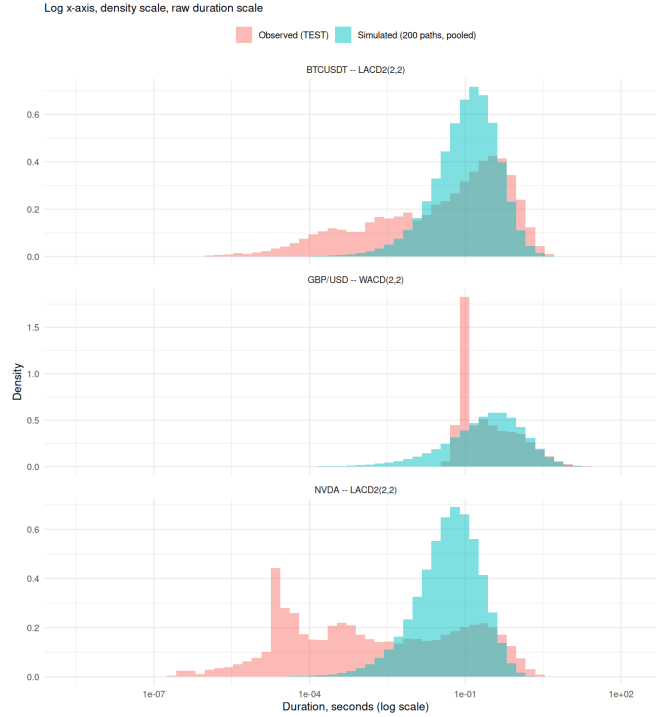


Figure 4.1: Observed and simulated test durations under the selected models, logarithmic scale

Figure 4.1 shows where that distance comes from. The simulated distributions sit in a narrow band around a tenth of a second for all three assets, while the observed series run several orders of magnitude wider. For BTCUSDT and NVDA the whole left tail is missing: the observed mass between 10^{-7} and 10^{-2} seconds, including the sharp NVDA peak near 10^{-5} , has no simulated counterpart at all. GBP/USD is closer, and the two overlap across most of the range, but its tall observed spike near 0.1 seconds reflects the 50 millisecond quantisation of that feed, whose minimum duration in Table 2.1 is 0.0500 seconds, rather than anything the duration process produces. No model reproduces it, and none should be expected to. The same comparison for the Poisson baseline is in Appendix B.2.

The autocorrelation results differ, and Figure 4.2 shows why. The observed autocorrelation decays with lag for every asset, from about 0.28 to 0.15 for BTCUSDT, 0.18 to 0.12 for GBP/USD and 0.17 to 0.09 for NVDA. The ACD specifications reproduce that shape; the baseline cannot, its curve being flat as independent draws require. It is not at zero, though, sitting near 0.055, 0.186 and 0.19, which is an effect of the rescaling in Equation (4.3): multiplying independent innovations by the diurnal factor imposes a common time-of-day pattern on every

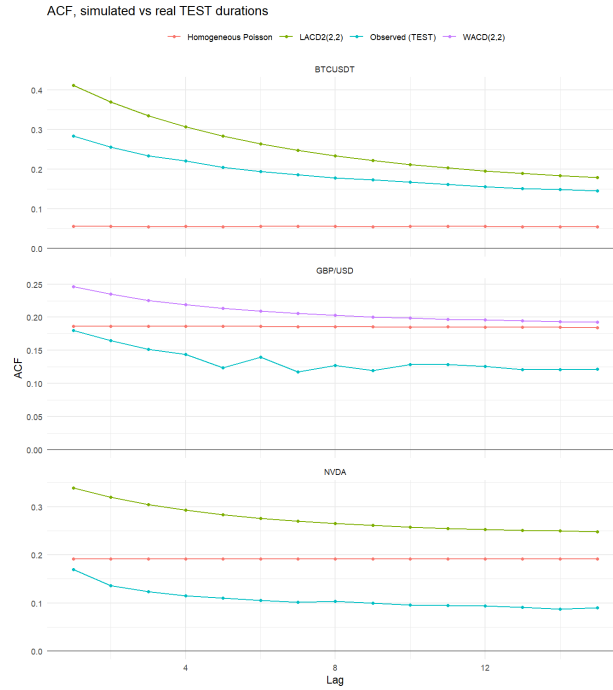


Figure 4.2: Autocorrelation of simulated and observed test durations, lags 1 to 15

path, and that pattern is itself autocorrelated. The flat line measures seasonality left in the raw scale, not captured dependence. Its height tracks the strength of the daily pattern in each asset: 0.186 for GBP/USD and 0.19 for NVDA, whose Ljung–Box statistics fell by 50.6% and 53.7% under the diurnal adjustment, against only 0.055 for BTCUSDT, whose fell by 4.5%. That ordering is what the explanation predicts, and confirms the flat line is seasonality rather than an error in the simulation.

The ACD specifications overshoot instead: at lag one the simulated autocorrelation is about 0.41 for BTCUSDT against 0.28 observed, 0.25 for GBP/USD against 0.18 and 0.34 for NVDA against 0.17, with the gap persisting across all fifteen lags. A model estimated at persistence just below one, run forward freely, carries shocks longer than the real series does.

This produces the split in Table 4.1. LACD2 clearly beats the baseline for BTCUSDT, 0.0654 against 0.1356, because the baseline’s flat autocorrelation sits far below the observed curve. For GBP/USD and NVDA the opposite happens: the flat baseline curve happens to sit near the average level of the observed decay, so the Poisson process records the smaller distance, winning by coincidence of height rather than by capturing any dependence. The ACD models capture the shape of the dependence but overstate its strength; the baseline captures no shape at all. Neither specification generates the very short durations that dominate BTCUSDT and NVDA, the same limitation Section 3.5 found from the other direction, so the result is mixed. The selected models beat a homogeneous Poisson process on distribution shape for every asset and on dependence for BTCUSDT, but not on dependence for GBP/USD or NVDA.

Chapter 5

Conclusion

5.1 Summary of findings

This dissertation explored how the timing between market trades is affected by recent activity, and whether we can use one unified model across different asset classes. It is observed that the durations in all three series show strong dependence, and it survives both removing the daily trading pattern and fitting the models. The second is mixed, since no single model wins everywhere.

The homogeneous Poisson process is rejected for every asset, with coefficients of variation of 1.9827, 1.6355 and 2.4317 against the one it requires. Removing the time-of-day pattern helps only where one exists: the Ljung–Box statistic falls by 50.6% for GBP/USD and 53.7% for NVDA but only 4.5% for BTCUSDT. Substantial dependence remains in all three adjusted series.

5.2 Model choice for each asset

For GBP/USD, WACD(2,2) is preferred: lowest residual statistics, lowest AIC and BIC, and a persistence of 0.9993, just inside the stationary region.

For BTCUSDT and NVDA, LACD2(2,2) is preferred. EACD(2,2) fits better in sample for both, with squared-residual Ljung–Box values of 0.2 and 0.8 against 4.2 and 5.4, but its persistence is above one, at 1.0022 and 1.0029, so it has no finite long-run mean and cannot generate a series. LACD2 is the best-fitting model among those that can, is stationary on the log scale, and handles sub-millisecond durations better.

5.3 Contribution

The main finding is that the above-one persistence for BTCUSDT and NVDA is mostly a product of how near-simultaneous trades are recorded, not of the duration process itself. Two checks agree. Merging events closer than a 10 ms cut-off lowers persistence from 1.0367 to 0.9943 for

BTCUSDT and from 1.0102 to 0.9950 for NVDA. Reducing sample size without merging does not, leaving it between 1.0297 and 1.0374 and between 1.0087 and 1.0136. Rebuilding events from the exchange record gives 1.0085 and 1.0010 (Appendix B.3).

A persistence estimate above one should therefore not be read as an economic finding until the event construction has been checked. GBP/USD supports this from the other side: it is the only asset with no sub-millisecond durations, and the only one whose persistence stays below one at every order under both linear specifications.

5.4 Limitations

The sample covers 1–23 May 2026, a single period that may reflect conditions specific to those weeks. The diagnostic tests are over-powered at these sample sizes: with 906,431 to 4,535,411 observations a Ljung–Box test rejects at any conventional level even for autocorrelations too small to matter, so they are better read as relative rankings than as pass or fail results. The GBP/USD diurnal spline is rank deficient, so that fit is not fully identified.

The Weibull cannot be used for BTCUSDT and NVDA: at shape estimates near 0.53 and 0.30 the fitted density diverges at zero and describes that divergence rather than duration behaviour. Persistence is measured on different scales in the linear and log families, so every stationarity claim rests on the linear EACD and WACD results, and the simulation compares average behaviour over 200 paths against a single test period.

5.5 Further work

The simulation could be strengthened with measures that compare whole distributions rather than percentile curves, such as a Kolmogorov–Smirnov statistic or a Wasserstein distance, and with interval estimates across paths rather than averages alone.

A second step is a distribution with a more flexible hazard shape. The Burr of Grammig and Maurer [8], the generalised gamma of Lunde [10] and the q-Weibull of Vuorenmaa [15] were built for the problem the Weibull runs into here, and the zero-inflated approach of Blasques et al. [3] handles sub-millisecond durations directly rather than as a nuisance. A third is to treat event construction as a modelling choice in its own right: it moves persistence more than any change of specification does, so reporting results under several event definitions would make such studies easier to compare.

Appendix A

Code

The code for this project is available at <https://github.com/Deepika-2004/Dissertation>.

Appendix B

Supplementary Figures and Tables

B.1 The ARMA(1,1) Model

The rejection of the two Poisson specifications establishes that durations are dependent. The natural first response is a standard linear time-series model for the adjusted series,

$$\tilde{x}_i = c + \phi \tilde{x}_{i-1} + \theta e_{i-1} + e_i, \quad e_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (\text{B.1})$$

where $\tilde{x}_i$ is the i th adjusted duration, c an intercept, ϕ the autoregressive coefficient governing dependence on the previous duration, θ the moving-average coefficient governing dependence on the previous errors, and e_i a Gaussian innovation of constant variance σ^2 .

This specification is not arbitrary. Engle and Russell [7] show that the autoregressive conditional duration recursion, admits an exact ARMA representation. Writing ψ_i for the conditional expectation of x_i given the history of the process, and defining

$$\eta_i = x_i - \psi_i, \quad (\text{B.2})$$

which is a martingale difference sequence by construction, substitution into the ACD recursion yields

$$x_i = \omega + (\alpha + \beta)x_{i-1} - \beta\eta_{i-1} + \eta_i. \quad (\text{B.3})$$

Here ω , α and β are the parameters of the ACD recursion defined in Equation (3.6): ω is a constant, α the coefficient on the previous duration, and β that on the previous conditional mean. Equation (B.3) has the algebraic form of Equation (B.1), with

$$\phi = \alpha + \beta, \quad \theta = -\beta. \quad (\text{B.4})$$

An ARMA(1,1) fit therefore delivers estimates of the ACD parameters without any specialised estimation, and the autoregressive coefficient ϕ is directly interpretable as the persistence $\alpha + \beta$ of the process.

The two specifications are nonetheless distinct, and the difference lies in the innovation. Equation (B.1) assumes e_i to be Gaussian, additive, and of constant variance. The ACD innovation of Equation (B.2) satisfies $\eta_i = \psi_i(\varepsilon_i - 1)$, where ε_i is the positive unit-mean innovation of Equation (3.5); its variance is therefore $\psi_i^2 \text{Var}(\varepsilon_i)$, which depends on the conditional mean. The ARMA specification is thus a Gaussian approximation to a process whose innovations are neither Gaussian nor homoskedastic.

Two consequences are testable. First, a Gaussian innovation is unbounded below, so a process of the form (B.1) assigns positive probability to negative durations; the magnitude of this probability is checked directly from the fitted series and reported with the ARMA results below. Second, the assumption of constant σ^2 implies that no dependence should remain in the squared residuals.

Estimation is by conditional sum of squares, which is asymptotically equivalent to full maximum likelihood and substantially faster at the sample sizes used here.

Results

Table B.1 reports the ARMA(1,1) fits. Each uses the first 100,000 adjusted training durations rather than the full sample, which is enough for the coefficients to settle and keeps the estimation comparable across the three assets.

Table B.1: ARMA(1,1) estimates on adjusted training durations, first 100,000 observations

Asset	$\hat{\phi}$	$\hat{\theta}$	$\hat{\alpha}$ implied	$\hat{\beta}$ implied	$\hat{\sigma}^2$	Mean
GBP/USD	0.9939	-0.9686	0.0253	0.9686	3.5920	0.9503
BTCUSDT	0.9634	-0.8550	0.1084	0.8550	2.5826	1.0712
NVDA	0.9711	-0.9249	0.0462	0.9249	6.8625	1.1363

The autoregressive coefficient sits close to unity for all three assets, at 0.9939, 0.9634 and 0.9711. Since $\hat{\phi}$ estimates $\alpha + \beta$ through the mapping in Equation (B.4), this recovers the near-unit persistence of Chapter 3 without any specialised estimation, which is a useful check that the result is a property of the data rather than of the ACD likelihood.

The split between the two implied coefficients is also informative. The implied β is large, between 0.8550 and 0.9686, while the implied α is small, between 0.0253 and 0.1084. The conditional mean therefore responds only weakly to any individual duration but carries almost all of its previous value forward, and it is that combination which produces persistence of this size. The same pattern appears in the EACD estimates of Table 3.3, where the first-order fits pair a small $\hat{\alpha}_1$ with a $\hat{\beta}_1$ close to one.

The fitted error variances show the cost of the Gaussian assumption. With $\hat{\sigma}^2$ of 3.5920, 2.5826 and 6.8625, the innovation standard deviations are 1.90, 1.61 and 2.62 against means near one. A Gaussian centred at the fitted mean with that spread places roughly 31%, 25%

and 33% of its mass below zero, so a process of the form Equation (B.1) assigns substantial probability to negative durations. This is the first of the two consequences noted above, and it is why the ARMA representation is useful for reading off persistence but not for describing durations themselves. The ACD specifications of Chapter 3 keep the same conditional-mean dynamics while replacing the additive Gaussian error with a positive multiplicative one, which removes the problem by construction.

B.2 Simulated and observed durations under the Poisson baseline

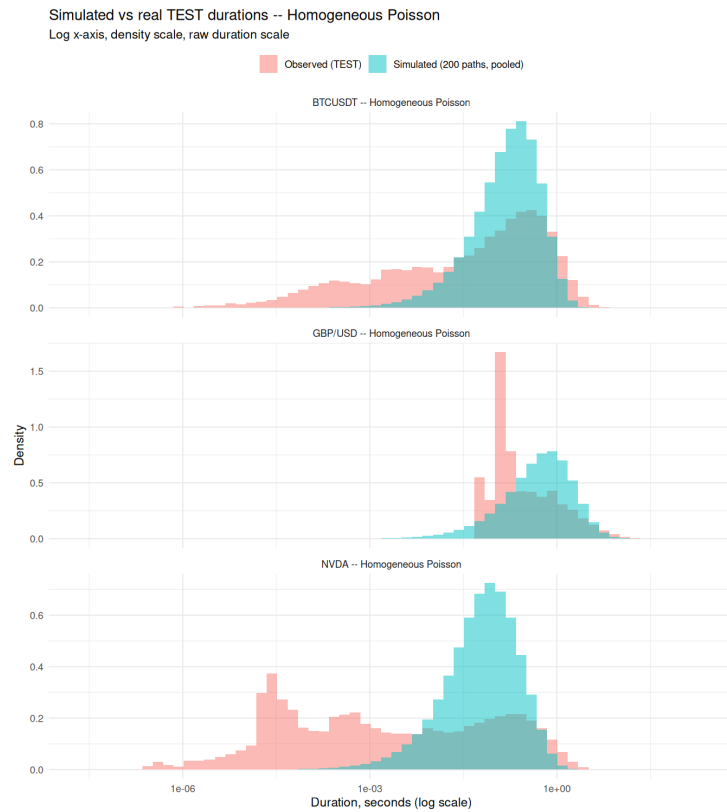


Figure B.1: Observed and simulated test durations under the homogeneous Poisson baseline, logarithmic scale

Figure B.1 repeats the comparison of Figure 4.1 for the homogeneous Poisson baseline. The simulated distribution is narrower still, and for GBP/USD it sits to the right of the observed mode rather than on top of it, which is what the quantile distance of 0.3177 in Table 4.1 records. For BTCUSDT and NVDA the picture is the same as for the selected models: the observed left tail has no simulated counterpart. Neither the baseline nor the ACD specifications generate durations near zero, so the improvement the selected models make is in the placement and spread of the bulk, not in the tail.

B.3 Reconstructing trades using exchange information

This check supports the persistence discussion in Section 3.4. Rather than using a time cut-off chosen during the analysis, it rebuilds the events using information supplied by the data source itself.

Table B.2: EACD(1,1) persistence after reconstructing trades using exchange-reported event information

Asset	Reconstruction method	Persistence
BTCUSDT	Merge consecutive same-side trades within 1 ms	1.0085
NVDA	Keep the exchange-flagged final print of each matching event	1.0010

The earlier check uses a time cut-off chosen by the analysis. This further check uses information supplied by the data source to identify records that belong to the same event. For BTCUSDT, consecutive same-side trades within 1 ms are combined. For NVDA, only the final print identified by the exchange's F.LAST flag is retained. Table B.2 shows persistence of 1.0085 for BTCUSDT and 1.0010 for NVDA. These values are much closer to one than the raw EACD(1,1) estimates of 1.0367 and 1.0102. Together with the merging and sample-size checks, this supports the view that separately recorded, closely spaced prints contribute to the above-one persistence estimates.

B.4 Characteristic roots of the second-order log-scale recursions

This table supports the stationarity discussion in Section 3.6.1, where several fitted second-order models carry a negative $\hat{\beta}_2$ and the coefficient sum on its own is not enough to decide whether the recursion is stationary.

Table B.3: Characteristic roots of the second-order log-scale recursions

Model	Asset	Order	$\hat{\beta}_1$	$\hat{\beta}_2$	z_1	z_2	$\sum \hat{\beta}_k$	Stationary
LACD1	GBP/USD	(2,2)	1.7293	-0.7290	0.9989	1.3733	1.0003	No
LACD1	BTCUSDT	(2,2)	1.7004	-0.7020	1.0055	1.4168	0.9984	Yes
LACD1	NVDA	(2,2)	1.5266	-0.5372	1.0241	1.8176	0.9894	Yes
LACD2	GBP/USD	(2,2)	1.8290	-0.8294	1.0024	1.2028	0.9996	Yes
LACD2	BTCUSDT	(2,2)	1.8719	-0.8721	1.0016	1.1448	0.9998	Yes
LACD2	NVDA	(2,2)	1.7901	-0.7915	1.0069	1.2547	0.9986	Yes

Table B.3 reports the roots z_1 and z_2 of Equation (3.29) for the second-order fits, where $\hat{\beta}_2$ is negative and the coefficient sum on its own does not settle the question. Both roots are real in

every case. The process is stationary when both lie outside the unit circle, so the smaller root is the one that matters.

For LACD2 the smaller root is 1.0024 for GBP/USD, 1.0016 for BTCUSDT and 1.0069 for NVDA. All three lie outside the unit circle, so the LACD2(2,2) recursions are stationary on the log scale, which the coefficient sums of 0.9996, 0.9998 and 0.9986 had already suggested. The margin is small in each case, and the fitted models sit close to the boundary rather than comfortably inside it.

LACD1 gives a mixed result. BTCUSDT and NVDA are stationary, with smaller roots of 1.0055 and 1.0241. GBP/USD is not: its smaller root is 0.9989, which lies inside the unit circle, matching its coefficient sum of 1.0003. The LACD1(2,2) fit selected for GBP/USD on residual diagnostics is therefore not stationary on the log scale, which is a further reason for not carrying LACD1 forward for that asset.

This check is why the coefficient sum alone is not enough. For LACD1 on GBP/USD the sum is 1.0003 and the smaller root is 0.9989, so both point the same way. For LACD2 on BTCUSDT the sum is 0.9998 and the smaller root is 1.0016, again agreeing. The two measures happen to agree here, but they are not the same test, and with a negative $\hat{\beta}_2$ only the roots settle it.

Bibliography

- [1] Bauwens, L., & Giot, P. (2000). The logarithmic ACD model: an application to the bid-ask quote process of three NYSE stocks. *Annales d'Économie et de Statistique* **60**, 117–149.
- [2] Belfrage, M. (2015). *ACDm: Tools for Autoregressive Conditional Duration Models*. R package version 1.0.4.
- [3] Blasques, F., Holý, V., & Tomanová, P. (2018). Zero-inflated autoregressive conditional duration model for discrete trade durations with excessive zeros. *arXiv preprint* arXiv:1812.07318.
- [4] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* **31**, 307–327.
- [5] Cavaliere, G., Mikosch, T., Rahbek, A., & Vilandt, F. (2025). Tail behavior and the integrated autoregressive conditional duration model. *arXiv preprint* arXiv:2505.06190.
- [6] Engle, R. F., & Russell, J. R. (1997). Forecasting the frequency of changes in quoted foreign exchange prices with the autoregressive conditional duration model. *Journal of Empirical Finance* **4**(2–3), 187–212.
- [7] Engle, R. F., & Russell, J. R. (1998). Autoregressive conditional duration: a new model for irregularly spaced transaction data. *Econometrica* **66**(5), 1127–1162.
- [8] Grammig, J., & Maurer, K.-O. (2000). Non-monotonic hazard functions and the autoregressive conditional duration model. *Econometrics Journal* **3**, 16–38.
- [9] Jasiak, J. (1998). Persistence in intertrade durations. Working paper, York University, Department of Economics.
- [10] Lunde, A. (1999). A generalized gamma autoregressive conditional duration model. Working paper, Aalborg University.
- [11] Nakakita, M., & Nakatsuma, T. (2025). Analysis of the trading interval duration for the Bitcoin market using high-frequency transaction data. *Quantitative Finance and Economics*, **9**(1), 202–241.

- [12] Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: a new approach. *Econometrica* **59**(2), 347–370.
- [13] Pacurar, M. (2008). Autoregressive conditional duration models in finance: a survey of the theoretical and empirical literature. *Journal of Economic Surveys* **22**(4), 711–751.
- [14] Tan, Y. F., Ng, K. H., Koh, Y. B., & Peiris, S. (2022). Modelling trade durations using dynamic logarithmic component ACD model with extended generalised inverse Gaussian distribution. *Mathematics*, 10(10), 1621.
- [15] Vuorenmaa, T. A. (2011). A q-Weibull autoregressive conditional duration model with an application to NYSE and HSE data. Available at SSRN 1952550.
- [16] Yan, X. (2021). *Autoregressive conditional duration modelling of high frequency data*. PhD thesis, University of Sussex.